

COMPUTER SCIENCE PROJECT

NEWS APPLICATION



SUBMITTED BY:

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NEWS SCOOPS – NEWS APPLICATION

(Using Python, Tkinter and SQLite)

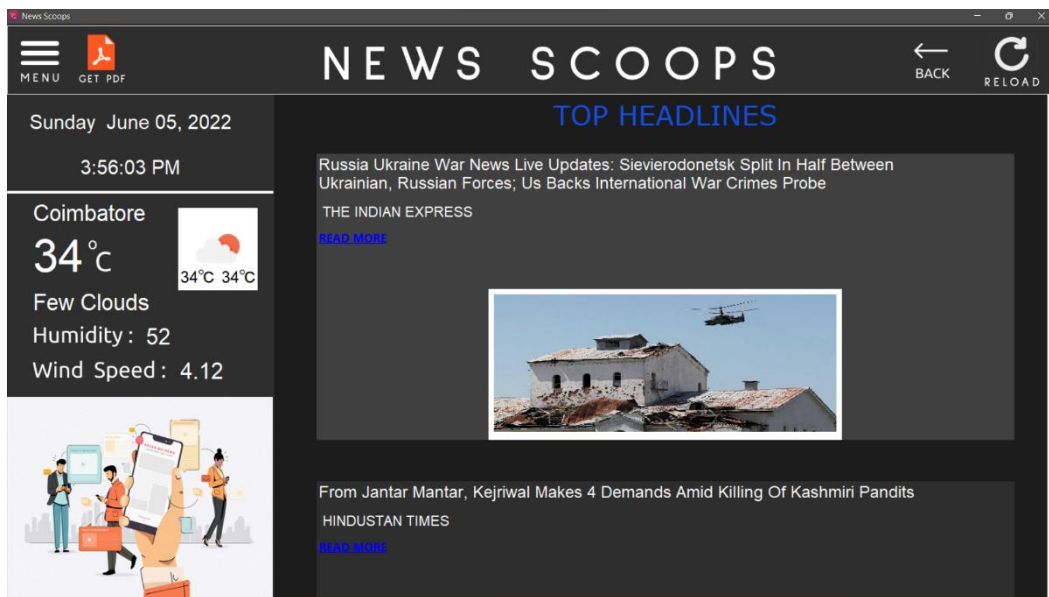


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SYNOPSIS

SYNOPSIS

The '**NEWS SCOOPS - NEWS APP**' is an application designed for getting news updates from various countries. This application will give the latest news updates at your fingertips. It has special features for users to register themselves easily. This application provides live news updates and also gives a preview of the news.

It redirects to the news source website from which the news is web scraped by the API. If you click on the news, you can get a preview of the news. You can view news of different types like Top Headlines, Sports, Technology, Science, Business, Health and Entertainment. This application is highly user friendly making it easy to use.

This application is designed using **Python 3.10.1(64 bits)** as the frontend. **VS Code (1.63.2)** is used as source code editor. VS code is very easy to use code editor which supports many languages and has debugging tools, syntax, it has many extensions for easy coding, highlighting etc. It has inbuilt features to help developers focus on the code without worrying about semantic errors, and leave syntax correction and indentation to the code editor. The Tkinter module is used for designing the framework of this news application.

SQLite is used as the backend for the application. SQLite is a C-language library that implements a small, fast, self-contained, high-reliability, full-featured, SQL database engine. SQLite is the most used database engine in the world. SQLite is built into all mobile phones and most computers and comes bundled inside countless other applications that people use every day.

INTRODUCTION

INTRODUCTION TO THE PROJECT

The ‘**NEWS SCOOPS – NEWS APPLICATION**’ is an application designed for users to view news of different categories. This application is user friendly and ensures easy viewing of news updates and also enables the user to go to the news source for getting more information. This application starts with login window, after logging in, it redirects to the main window.

MAIN WINDOW:

This application is accessed through a main window. The main window has 4 sections.

- The Title Bar
- The News Pane
- The Menu
- The Weather and Time Pane

THE TITLE BAR:

This Bar consists of a label which consists of an image which has the title News Scoops. This also consists of 4 buttons.

- ❖ **Menu Button:** This button moves the menu
- ❖ **Reload Button:** This button reloads the news content
- ❖ **Back Button:** This button redirects you to previous news you were viewing.
- ❖ **GetPDF Button:** This button generates a pdf of the top headlines at that moment and you can save it in your device.

THE NEWS PANE:

It is a scrollable window which consists of frames according to the number of news acquired by the API. These frames contain the Headlines, the Publisher, the Read More HTML LABEL which directs to the news source website, it also contains an image related to the news which is shown through a HTML LABEL.

THE MENU:

This is a frame which moves in or out the application as and when you click the menu button. The menu consists of 8 buttons i.e., Top Headlines, Technology, Health, Science, Sports, Entertainment which will show the respective category of news in the news pane and it also consists of the Log Out Button to log out of the application.

THE WEATHER AND TIME PANE:

This is a frame which shows the weather and time. In the weather pane it shows the temperature (also min and max), description of Weather, picture of the weather, Place, Humidity and the wind speed. In the time pane it shows the time, date and the day.

The main feature of News Scoops is the timely updating news where you can have a preview of the news as well as you can directly go to the news source for more details regarding the news. This app makes use of the requests module to connect to the internet whenever required and makes requests to the NewsApi's API to get news data. The API sends JSON response which is parsed and saved in the SQLite database and it is then displayed in the news application.

SYSTEM CONFIGURATION

SYSTEM CONFIGURATION

SOFTWARE USED:

- Python 3.10.1 (64-bit): – Front End
- VS Code (1.63.2): – Source Code Editor
- Microsoft® Windows 11: – Operating System
- SQLite: – Back End

HARDWARE SPECIFICATIONS:

- 16 GB RAM – RAM
- 1 TB SSD – Storage
- Intel® CORE™ i7 – Processor
- 1920x1080 Monitor – Output Device
- Touchpad, Keyboard – Input Device

ABOUT PYTHON

ABOUT PYTHON

Python is a high-level, general-purpose and a very popular programming language. Python allows programming in Object-Oriented and Procedural paradigms. Its language constructs an approach which aims to help programmers write clear, logical code for small and large-scale projects. Python was created in the late 1980s, and first released in 1991 by Guido Van Rossum. It is based on the languages ABC and Modula3.

ADVANTAGES:

- Versatile, Easy to use
- Interpreted language
- Free and Open Source with Vibrant Community
- Free learning resources

DISADVANTAGES:

- Python has high memory consumption
- Python is slow at runtime
- It has much lesser libraries than other languages
- It is difficult to convert into other languages

MODULES USED

MODULES USED

❖ **TKINTER:**

Tkinter is the standard GUI library for Python. Python when combined with Tkinter provides a fast and easy way to create GUI applications.

❖ **PIL:**

Python Imaging Library is a free and open-source additional library for the Python programming language that adds support for opening, manipulating, and saving many different images file formats.

❖ **SQLITE3:**

Python SQLite3 module is used to integrate the SQLite database with Python. It provides a straightforward and simple-to-use interface for interacting with SQLite databases.

❖ **TIME:**

Time module is a module in Python which is used to provide various time related functions.

❖ **REQUESTS:**

The requests module allows you to send HTTP requests using Python. The HTTP request returns a Response Object with all the response data (content, encoding, status, etc).

❖ **TEXT WRAPPER:**

The text wrap module can be used for wrapping and formatting of plain text. This module provides formatting of text by adjusting the line breaks in the input paragraph.

❖ **TKINTER WEB:**

A fast and lightweight cross-platform web browser widget for Tkinter. It which enables loading HTML and CSS code into Tkinter applications.

❖ **TKHTMLVIEW:**

This module is a collection of Tkinter widgets whose text can be set in HTML format. A HTML widget isn't a web browser frame, it's only a simple and lightweight HTML parser that formats the tags used by the Tkinter text base class.

❖ **PANDAS:**

This is a software library written for the Python programming language for data manipulation and analysis. In particular, it offers data structures and operations for manipulating numerical tables and time series.

❖ **ITERTOOLS:**

Itertools is a module in Python, it is used to iterate over data structures that can be stepped over using a for-loop. Such data structures are also known as iterables. This module works as a fast, memory-efficient tool that is used either by themselves or in combination to form iterator algebra.

❖ **FPDF:**

FPDF is a library for PDF document generation under Python. It helps in easy generation of PDF files. It has many tools for creating PDF in a creative way.

❖ **UNICODEDATA:**

This module provides access to the Unicode Character Database (UCD) which defines character properties for all Unicode characters.

TKINTER MODULE

TKINTER MODULE

Tkinter is the standard GUI library for Python. Python when combined with Tkinter provides a fast and easy way to create GUI applications. Tkinter provides a powerful object-oriented interface to the Tk GUI toolkit.

Here, objects called widgets make up the window and have master or parent widget. Properties of the widgets are specified with keyword arguments. Keyword arguments have the same name as the corresponding resource under Tk.

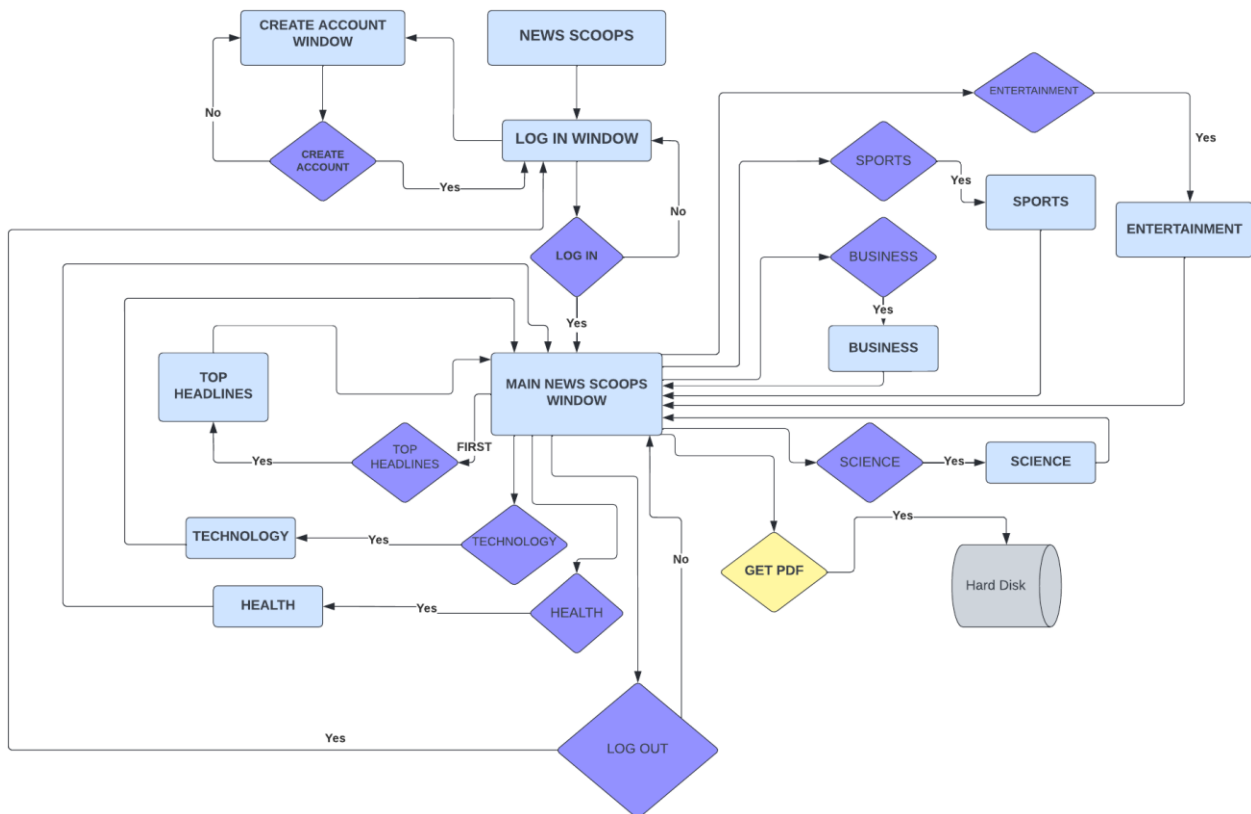
Widgets are positioned with one of the geometry managers Place, Pack. These managers can be called with methods place, pack, available in every Widget.

The various widgets used in this application are:

- **Button:** It is like a label on clicking which a function is called
- **Canvas:** It is used to display various graphics on the screen.
- **Entry:** It is used to accept single line text from user.
- **Frame:** It is a container of a window where other widgets can be added
- **Scrollbar:** It is a slide scroller that is used to move the contents in a screen.
- **Label:** It is a display box where text or images are placed

DATA FLOW DIAGRAM

DATAFLOW DIAGRAM



A Data Flow Diagram (DFD) maps out the flow of information for any process or system. It uses defined symbols like rectangles, circles and arrows, plus short text labels, to show data inputs, outputs, storage points and the routes between each destination. This Dataflow Diagram shows the flow of data of the news application. It gives an idea of the overall news application.

SQLITE

SQLITE

SQLite is a relational database management system (RDBMS) contained in a C library. In contrast to many other database management systems, SQLite is not a client–server database engine. Rather, it is embedded into the end program. It is the most widely deployed database engine, as it is used by several of the top web browsers, operating systems, mobile phones, and other embedded systems. SQLite is very carefully tested prior to every release and has a reputation for being very reliable. Most of the SQLite source code is devoted purely to testing and verification. The database file format is cross-platform - you can freely copy a database between 32-bit and 64-bit systems or between big-endian and little-endian architectures.

ADVANTAGES:

- Flexibility of open source
- Better performance
- No installation needed
- Portable and Accessible

DISADVANTAGES:

- Lack of multi-user capabilities
- Lack of friendly user management system
- Lack of security capabilities beyond encrypting the database file itself.

TABLES CREATED

During the execution of the news application two tables are created in SQLite.

TABLE – (USER ACCOUNTS):

(PRIMARY KEY: ID)

cid	name	type	notnull	dflt_value	pk
0	id	integer	0	NULL	1
1	name	text	1	NULL	0
2	email	text	1	NULL	0
3	password	text	1	NULL	0
4	phno	integer	1	NULL	0

The above table is used to store details of the user like Name, Email, Password, Phone Number as and when the user creates an account. The ID of the user is auto incrementing.

TABLE – (NEWS DATABASE):

cid	name	type	notnull	dflt_value	pk
0	index	INTEGER	0	NULL	0
1	title	TEXT	0	NULL	0
2	url	TEXT	0	NULL	0
3	urlToImage	TEXT	0	NULL	0
4	content	TEXT	0	NULL	0

The above table is used to store the data parsed from the JSON given by the API as a response object. It stores the titles, URL, URL to the image and the content of the news.

SOURCE CODE

SOURCE CODE

Importing Required Modules

```
import time,requests,json
from tkinter import messagebox
from tkinter import *
import tkinter.filedialog as filedialog
from tkinter import ttk
from PIL import ImageTk,Image
from tkinterweb.htmlwidgets import HtmlLabel
import textwrap
from tkhtmlview import HTMLLabel
import sqlite3
from datetime import datetime,date
import pandas
import itertools
from fpdf import FPDF
import unicodedata
```

Setting Window Configurations

```
window = Tk()
window.title('News Scoops')
window.overrideredirect(True)
w,h = window.winfo_screenwidth(),window.winfo_screenheight()
app_width,app_height=1280,720
x=(w/2)-(app_width/2)
y=(h/2)-(app_height/2)
window.geometry(f'{app_width}x{app_height}+{int(x)}+{int(y)}')
window.iconphoto(False,ImageTk.PhotoImage(file='pics/News Scoop trans.png'))
```

Setting Theme

```
window.tk.call('source','Sun-Valley-ttk-theme-master/sun-valley.tcl')
window.tk.call("set_theme", "dark")
```

Initialising Account Database

```
conector=sqlite3.connect('news_scoops_user_accounts.db')
cursor=conector.cursor()
cursor.execute('CREATE TABLE IF NOT EXISTS user_details(indexes integer PRIMARY KEY
AUTOINCREMENT,name text NOT NULL,email text NOT NULL,password text NOT NULL,phno integer NOT
NULL)')
conector.commit()
conector.close()
```

```
news_gen_connector=sqlite3.connect('news_scoops_news_gen.db')
news_gen_cursor=news_gen_connector.cursor()
news_gen_cursor.execute('CREATE TABLE IF NOT EXISTS news_gen( id integer PRIMARY KEY
AUTOINCREMENT, title text NOT NULL, url text NOT NULL , urltoimg text, content text )')
news_gen_connector.commit()
news_gen_connector.close()
```

Defining Functions

```
def log_in_main_window():
    global in_which,where_list,menu_frame_info
    conector=sqlite3.connect('news_scoops_user_accounts.db')
    cursor=conector.cursor()
    email=email_id_textbox.get()
```



```

passw=password_textbox.get()

if email=="and passw==" :
    messagebox.showwarning('News Scoops','Enter Both Email Password !')
elif email==" or passw==" :
    messagebox.showwarning('News Scoops','Enter Both Email Password !')
if email!=" and passw!=":
    cursor.execute(f"SELECT COUNT(*) from user_details WHERE email = '{email}'")
    result=cursor.fetchone()

    if int(result[0])==0:
        messagebox.showwarning('News Scoops','This Username Does Not Exist !')
        email_id_textbox.delete(0,END)
        password_textbox.delete(0,END)

    else:
        cursor.execute(f"SELECT * from user_details WHERE email='{email}'")
        res=cursor.fetchone()
        if res[3] == passw:
            conector.commit()
            conector.close()
            email_id_textbox.delete(0,END)
            password_textbox.delete(0,END)
            log_in_window_label.forget()
            email_id_textbox.forget()
            password_textbox.forget()
            log_in_button.forget()
            create_account_button.forget()
            in_which='top headlines'
            where_list=['top headlines']
            create_title_bar()
            create_weather_time_window()
            menu_frame_info='i-o'
            menu_place()
            populate(in_which)

        else:
            messagebox.showwarning('News Scoops','Wrong Password ! Please Enter Correct Password !')
            password_textbox.delete(0,END)

def reload():# Destroying The Inner Elements Then Proceeding
    news_frame1.destroy()
    populate(in_which)
    if menu_frame_info=='o-i':
        menu_mover()

def menu_mover():
    global menu_frame_info
    if menu_frame.wininfo_x() == -305:
        for i in [-305,-200,-100,-20,0]:
            menu_frame.configure(menu_frame.place(x=i,y=103))
            window.update()
            time.sleep(0.01)
            menu_frame_info='o-i'
    else:
        for i in [0,-20,-100,-200,-305]:
            menu_frame.configure(menu_frame.place(x=i,y=103))
            window.update()
            time.sleep(0.01)
            menu_frame_info='i-o'

def getweather():
    global weatherdata,cityname

```

```

weather_api_key='069dd683a7928cef21845176b4f9aa5f'
cityname='Coimbatore'
weather_api_link=f'https://api.openweathermap.org/data/2.5/weather?q={cityname}&appid={weather_api_key}'
weatherdata=requests.get(weather_api_link).json()

news_pic_test_img=Image.open('pics\pic of news.png')
news_resized_pic_img=news_pic_test_img.resize((388,295),Image.ANTIALIAS)
news_pic_img=ImageTk.PhotoImage(image=news_resized_pic_img)

def weather_populate():
    getweather()
    if weatherdata['cod']=='404':
        error_label=Label(weather_time_frame)
        error_label.place(x=60,y=60)
    else:
        temp=int(round(weatherdata['main']['temp']-273.15,0))
        weather_desp=weatherdata['weather'][0]['description'].title()
        hmdt=weatherdata['main']['humidity']
        windspd=weatherdata['wind']['speed']
        weather_icon = weatherdata['weather'][0]['icon']
        temp_min=str(int(round(weatherdata['main']['temp_min']-273.15,0))) +chr(0x2103)
        temp_max=str(int(round(weatherdata['main']['temp_max']-273.15,0))) +chr(0x2103)
        cityname_label=Label(weather_time_frame,text=cityname,font=('Helvetica',25),background='#2e2e2e')
        temp_label=Label(weather_time_frame,text=temp,font=('Helvetica',50),background='#2e2e2e')
        weather_desp_label=Label(weather_time_frame,text=weather_desp,font=('Helvetica',25),background='#2e2e2e')
        hmdt_label=Label(weather_time_frame,text=hmdt,font=('Helvetica',25),background='#2e2e2e')
        windspd_label=Label(weather_time_frame,text=windspd,font=('Helvetica',25),background='#2e2e2e')
        news_pic_label=Label(weather_time_frame,image=news_pic_img,borderwidth=0)
        weather_icon_label=HtmlLabel(weather_time_frame)
        temp_min_label=Label(weather_time_frame,text=temp_min,font=('Helvetica',18),background='white',
        foreground='black')
        temp_max_label=Label(weather_time_frame,text=temp_max,font=('Helvetica',18),background='white',
        foreground='black')
        cityname_label.place(x=35,y=150)
        temp_label.place(x=35,y=195)
        weather_desp_label.place(x=35,y=280)
        hmdt_label.place(x=200,y=330)
        windspd_label.place(x=250,y=380)
        news_pic_label.place(x=0,y=440)
        weather_icon_label.load_html(f"<html><body><img src='http://openweathermap.org/img/wn/{weather_icon}.png' style='width:100px; height: 100px; object-fit: cover;'></body></html>")
        weather_icon_label.place(x=250,y=165)
        temp_min_label.place(x=250,y=250)
        temp_max_label.place(x=310,y=250)

def convert(s):
    return unicodedata.normalize('NFKD', s).encode('latin-1', 'ignore').decode('latin-1')

def get_pdf():
    savedialog= filedialog.asksaveasfile(defaulttextextension='.pdf',initialdir=""C:/Users/SimiLemi/Documents',
    title='Save File',initialfile=
    'NewsScoops Todays Headlines')
    pdf= FPDF()
    pdf.alias_nb_pages()
    pdf.add_page()
    pdf.set_font('Helvetica', "", 12)
    pdf.image('pics/News Scoop trans.png',x=85,y=10,w=40,h=40)
    pdf.ln(45)
    getnews('top headlines')
    for i in range(len(titles)):
        text=convert(titles[i])
        pdf.multi_cell(0,5,f'{i+1}. '+text.upper(),0,1)

```

```

    pdf.ln(5)
    pdf.output(savedialog.name)

def getnews(news_type):
    global titles,links_to_web,url_to_img,content_of_article
    country='in'
    api_key='69451af3dd2e48119efe3e9dcdd219c5'

    if news_type=='top headlines':
        url1=f'https://newsapi.org/v2/top-headlines?pageSize=40&country={country}&apiKey={api_key}'

    if news_type in ['business','entertainment','science','technology','sports','health']:
        url1=f'https://newsapi.org/v2/top-headlines?pageSize=40&category={news_type}&language=en
        &apiKey={api_key}'

    news1=requests.get(url1).json()
    articles1=news1['articles']
    for redarticle in articles1:
        del redarticle['source']
        del redarticle['author']
        del redarticle['description']
        del redarticle['publishedAt']

    df = pandas.json_normalize(articles1)
    news_gen_connector=sqlite3.connect('news_scoops_news_gen.db')
    news_gen_cursor = news_gen_connector.cursor()
    df.to_sql("news_gen",news_gen_connector,if_exists='replace')
    news_gen_cursor.execute('update news_gen set urlToImage="https://www.feednavigator.com/var/
    wrbm_gb_food_pharma/storage/images/9/2/8/5/235829-6-eng-GB/Feed-Test-SIC-Feed-20142.jpg"
    where urlToImage is null')
    news_gen_connector.commit()
    news_gen_cursor.execute('select * from news_gen')
    news_res=news_gen_cursor.fetchall()

    titles=[]
    links_to_web=[]
    url_to_img=[]
    content_of_article=[]

    for i in news_res :
        titles.append(i[1])
        links_to_web.append(i[2])
        url_to_img.append(i[3])
        content_of_article.append(i[4])

weather_pane_label_img=ImageTk.PhotoImage(file='pics\WEATHER PANE BLACK.png')

def create_weather_time_window():

    global time_label,weather_time_frame
    weather_time_frame=Frame(window,borderwidth=0)
    weather_time_label=Label(weather_time_frame,width=55,height=43,background='#2e2e2e')
    time_label=Label(weather_time_frame ,text="",background='#2e2e2e',font=('Helvetica',22))
    weather_pane_label=Label(weather_time_frame,image=weather_pane_label_img,borderwidth=0)
    weather_time_frame.place(x=0,y=103)
    weather_time_label.pack()
    time_label.place(x=30,y=20)
    weather_pane_label.place(x=0,y=140)
    clock()
    weather_populate()

```

```

def create_scroll_window(): #News Workspace
    global news_frame1,news_canvas,news_frame2,news_vsb
    news_frame1= Frame(window,borderwidth=0)
    news_canvas=Canvas(news_frame1, borderwidth=0)
    news_frame2= Frame(news_canvas,borderwidth=0)
    news_vsb=Scrollbar(news_canvas, orient="vertical", command=news_canvas.yview)
    news_canvas.configure(yscrollcommand=news_vsb.set)
    news_frame2.bind("<Configure>",lambda e:news_canvas.configure(scrollregion= news_canvas.bbox("all")))
    news_frame1.place(x=390, y=103, width=1146, height=735)
    news_frame2.pack(fill=BOTH,expand=1)
    news_vsb.pack(side="right", fill="y")
    news_canvas.pack(side="left", fill="both", expand=True)
    news_canvas.create_window((0,0), window=news_frame2, anchor="nw",tags="news_frame2")

def menu_forget():

    menu_frame.destroy()
    top_headlines_button.destroy()
    technology_button.destroy()
    health_button.destroy()
    science_button.destroy()
    business_button.destroy()
    sport_button.destroy()
    entertainment_button.destroy()
    log_out_button.destroy()

def menu_place():
    global menu_frame,top_headlines_button,business_button,entertainment_button,science_button,technology_button,
    sport_button,health_button,log_out_button
    menu_frame=Frame(window,background='#2e2e2e',borderwidth=0)
    top_headlines_button=Button(menu_frame,image=top_headlines_button_img,borderwidth=0,bd=0,
    highlightthickness=0,command=top_headlines)
    business_button=Button(menu_frame,image=business_button_img,borderwidth=0,bd=0,
    highlightthickness=0,command=business)
    entertainment_button=Button(menu_frame,image=entertainment_button_img,borderwidth=0,bd=0,
    highlightthickness=0,command=entertainment)
    science_button=Button(menu_frame,image=science_button_img,borderwidth=0,bd=0,highlightthickness=0,
    command=science)
    technology_button=Button(menu_frame,image=technology_button_img,borderwidth=0,bd=0,
    highlightthickness=0,command=technology)
    sport_button=Button(menu_frame,image=sport_button_img,borderwidth=0,bd=0,highlightthickness=0,
    command=sport)
    health_button=Button(menu_frame,image=health_button_img,borderwidth=0,bd=0,highlightthickness=0,
    command=health)
    log_out_button=Button(menu_frame,image=log_out_button_img,borderwidth=0,bd=0,highlightthickness=0,
    command=log_out)
    menu_frame.place(x=-305,y=103,width=305,height=740)
    top_headlines_button.pack()
    technology_button.pack()
    health_button.pack()
    science_button.pack()
    business_button.pack()
    sport_button.pack()
    entertainment_button.pack()
    log_out_button.pack()

def read_more(e):
    menu_forget()
    menu_button.place_forget()
    get_pdf_button.place_forget()
    back_button_title.place_forget()
    reload_button.place_forget()
    news_frame1.destroy()

```

```

def go_to_deatil_window():

    global detail_title_label,detail_image_label,detail_content_string,detail_html_link_label,detail_content_label
    create_scroll_window()
    detail_label=Label(news_frame2,text='PREVIEW OF NEWS',font=('Verdana',30),background='#1c1c1c',
    foreground='#154fe2')
    detail_label.pack()
    news_frame3=Frame(news_frame2,background='black')
    news_frame3.pack(fill=BOTH,expand=1,pady=10)
    back_button.place(x=10,y=20)

    detail_title_value=titles[e]
    detail_title_wrapper=textwrap.TextWrapper(width=75)
    detail_title_string=detail_title_wrapper.fill(text=detail_title_value)
    detail_title_label=Label(news_frame3,text=detail_title_string,font=('Helvetica',20,'bold'),background='black',
    justify=LEFT)
    detail_title_label.pack(anchor='nw',)

    detail_image_label=HtmlLabel(news_frame3)
    detail_image_label.load_html(f'<html><body>
    </body></html>',url_to_img[e])
    detail_image_label.pack(pady=30)
    detail_content_value=content_of_article[e]

    if detail_content_value==None:
        detail_content_value="

    else:

        detail_content_wrapper=textwrap.TextWrapper(width=70)
        detail_content_string=detail_content_wrapper.fill(text=detail_content_value)
        stuff_to_remove_content_string = ['<ul>','</ul>','<ol>','</ol>','</li>','<li>']
        for i in stuff_to_remove_content_string:
            detail_content_string.replace(i,"")

        detail_content_label=Label(news_frame3,text=detail_content_string,font=('Helvetica',20,'bold'),
        background='black',justify=LEFT)
        detail_content_label.pack(pady=30)

        detail_html_link=links_to_web[e]
        detail_html_link_label=HTMLLabel(news_frame3,html=f"<h2><a href='{detail_html_link}'>READ
        MORE</a></h2>",
        background='black',width=30,height=5)
        detail_html_link_label.pack(anchor='nw')
        empty_label=Label(news_frame3,width=160,height=5,background='black')
        empty_label.pack()

    go_to_deatil_window()

def back_button_cmd():

    menu_place()
    back_button.place_forget()
    menu_button.place(x=10,y=10)
    get_pdf_button.place(x=100,y=10)
    back_button_title.place(x=1310,y=10)
    reload_button.place(x=1425,y=10)
    news_frame1.destroy()
    populate(in_which)

ret=lambda e:(lambda p : read_more(e))

```

```

def populate(news_type):

    global newsshowerframe,news_empty_frame,htmllabels,title_label,author_label,htmllinks,which_news_label
    create_scroll_window()
    newsshowerframe={}
    news_empty_frame={}
    htmllabels={}
    htmllinks={}
    title_label={}
    author_label={}
    getnews(news_type)
    colour_count=0
    colour_list=[('#404040','white'),('#2e2e2e','white')]
    which_news_label=Label(news_frame2,text=in_which.upper(),font=('Verdana',30),foreground='#154fe2')
    which_news_label.pack(anchor='center')

    for i in range(len(titles)):

        if colour_count>=len(colour_list):
            colour_count=0
            newsshowerframe[i]=Frame(news_frame2,background=colour_list[colour_count][0])
            newsshowerframe[i].pack(fill=BOTH,expand=1,padx=60,pady=30,anchor='nw',ipadx=9)
            newsshowerframe[i].bind('<Button-1>',ret(i))
            news_empty_frame[i]=Frame(newsshowerframe[i],background=colour_list[colour_count][0])
            news_empty_frame[i].pack(fill=BOTH,expand=1,ipadx=16)
            news_empty_frame[i].bind('<Button-1>',ret(i))
            colour_count+=1

    colour_count=0

    for i in range(len(titles)):
        if colour_count>=len(colour_list):
            colour_count=0

            news_shower_text=list(str(titles[i]).rpartition('-'))
            if '-' in news_shower_text[0][-1:-4:-1]:
                temp_list=list(news_shower_text[0].rpartition('-'))
                news_shower_text[2]=temp_list[2]+' '+news_shower_text[2]
                news_shower_text[0]=temp_list[0]
            news_title_string=news_shower_text[0].title()
            news_author_text=news_shower_text[2].upper()
            news_title_text_wrapper=textwrap.TextWrapper(width=80)
            news_title_text=news_title_text_wrapper.fill(text=news_title_string)

            title_label[i]=Label(news_empty_frame[i],text=news_title_text,font=('Helvetica',18),background=colour_list
            [colour_count][0],justify=LEFT,
            fg=colour_list[colour_count][1])
            title_label[i].pack(anchor='nw')
            title_label[i].bind('<Button-1>',ret(i))

            author_label[i]=Label(news_empty_frame[i],text=f'{news_author_text}',font=('Helvetica',16),background=
            colour_list[colour_count][0],fg=colour_list[colour_count][1])
            author_label[i].pack(anchor='nw',pady=10)
            author_label[i].bind('<Button-1>',ret(i))

            colour_count+=1

    colour_count=0

    for i in range(len(titles)):
        if colour_count>=len(colour_list):
            colour_count=0
            htmllinks[i]=HTMLLabel(news_empty_frame[i],html=f'<h5><a href="{links_to_web[i]}">READ MORE </a></h5>

```

```

", background=colour_list[colour_count][0],width=20,height=5)
htmlmlinks[i].pack(anchor='nw')
colour_count+=1

```

```

for i in range(len(titles)) :
    htmlmlabels[i]=HtmlLabel(newsshowerframe[i])
    htmlmlabels[i].load_html(f<html><body><img src = "{url_to_img[i]}" width="500" height="200"/></body>
</html>',url_to_img[i])
    htmlmlabels[i].on_downloading_resource(htmlmlabels[i].pack)
    htmlmlabels[i].bind('<Button-1>',ret(i))

```

Defining More Functions

```
def create_account_navigation():
```

```

    email_id_textbox.delete(0,END)
    password_textbox.delete(0,END)
    log_in_window_label.place_forget()
    email_id_textbox.place_forget()
    password_textbox.place_forget()
    log_in_button.place_forget()
    create_account_button.place_forget()
    create_account_window_label.pack()
    create_account_window_create_account_button.place(x=730,y=720)
    create_account_back_button.place(x=400,y=722)
    create_account_name_textbox.place(x=550,y=350)
    create_account_email_textbox.place(x=550,y=440)
    create_account_password_textbox.place(x=550,y=530)
    create_account_phno_textbox.place(x=550,y=620)

```

```
def create_account_window_create_account():
```

```

    new_name = create_account_name_textbox.get()
    new_email=create_account_email_textbox.get()
    new_password=create_account_password_textbox.get()
    new_phno=create_account_phno_textbox.get()
    if new_name==" or new_email=="or new_password==" or new_phno=="":
        messagebox.showwarning('News Scoops','Dont leave any fields empty !')
    else:

```

```

        is_email=True
        is_phno=True
        is_password=True
        warning=""
        if '@' not in new_email:
            is_email=False
            warning += "\nInvalid Email"
        else:
            split_email=new_email.partition('@')
            if split_email[2] not in ['gmail.com','reddit.com','yahoo.com','yahoo.co.in','hotmail.com','outlook.com']:
                is_email=False
                warning += "\nInvalid Email"

```

```

        if len(new_phno)!=10:
            is_phno=False
            warning+= "\nInvalid Phno(Please Type 10-digit Phone Number)"
        if len(new_password)>20:
            is_password=False
            warning+= "\nInvalid Password (Password Must be within 20 characters)"
        if is_email==False or is_phno==False or is_password==False:
            messagebox.showwarning('News Scoops',warning)

```

```

        if is_email and is_phno and is_password:

```

```

conector=sqlite3.connect('news_scoops_user_accounts.db')
cursor=conector.cursor()
cursor.execute(f"SELECT COUNT(*) from user_details WHERE email = '{new_email}'")
result=cursor.fetchone()
print(result)

if int(result[0])>0:

    messagebox.showwarning('News Scoops','The username already exists')
    create_account_name_textbox.delete(0,END)
    create_account_email_textbox.delete(0,END)
    create_account_password_textbox.delete(0,END)
    create_account_phno_textbox.delete(0,END)

else:

    cursor.execute('INSERT INTO user_details(name,email,password,phno)VALUES(:name,:email,:
password,:phno)', {'name':new_name,'email':new_email,'password':new_password,'phno':new_phno})
    conector.commit()
    conector.close()

    create_account_name_textbox.delete(0,END)
    create_account_email_textbox.delete(0,END)
    create_account_password_textbox.delete(0,END)
    create_account_phno_textbox.delete(0,END)

    create_account_window_label.pack_forget()
    create_account_window_create_account_button.place_forget()
    create_account_back_button.place_forget()
    create_account_name_textbox.place_forget()
    create_account_email_textbox.place_forget()
    create_account_password_textbox.place_forget()
    create_account_phno_textbox.place_forget()
    log_in_window_label.place(x=0,y=0)
    email_id_textbox.place(x=500,y=400)
    password_textbox.place(x=500,y=540)
    log_in_button.place(x=600,y=625)
    create_account_button.place(x=760,y=735)

    messagebox.showinfo('News Scoops','You have created your News Scoops Account!')

def top_headlines():

    global in_which
    menu_mover()
    news_frame1.destroy()
    populate('top headlines')
    which_news_label.configure(text='TOP HEADLINES')
    in_which='top headlines'
    where_list.append(in_which)

def business():

    global in_which
    menu_mover()
    news_frame1.destroy()
    populate('business')
    which_news_label.configure(text='BUSINESS')
    in_which='business'
    where_list.append(in_which)

```



```

def entertainment():

    global in_which
    menu_mover()
    news_frame1.destroy()
    populate('entertainment')
    which_news_label.configure(text='ENTERTAINMENT')
    in_which='entertainment'
    where_list.append(in_which)

def science():

    global in_which
    menu_mover()
    news_frame1.destroy()
    populate('science')
    which_news_label.configure(text='SCIENCE')
    in_which='science'
    where_list.append(in_which)

def technology():

    global in_which
    menu_mover()
    news_frame1.destroy()
    populate('technology')
    which_news_label.configure(text='TECHNOLOGY')
    in_which='technology'
    where_list.append(in_which)

def sport():

    global in_which
    menu_mover()
    news_frame1.destroy()
    populate('sports')
    which_news_label.configure(text='SPORTS')
    in_which='sports'
    where_list.append(in_which)

def health():

    global in_which
    menu_mover()
    news_frame1.destroy()
    populate('health')
    which_news_label.configure(text='HEALTH')
    in_which='health'
    where_list.append(in_which)

def create_account_back():

    create_account_name_textbox.delete(0,END)
    create_account_email_textbox.delete(0,END)
    create_account_password_textbox.delete(0,END)
    create_account_phno_textbox.delete(0,END)
    create_account_window_label.pack_forget()
    create_account_window_create_account_button.place_forget()
    create_account_back_button.place_forget()
    create_account_name_textbox.place_forget()
    create_account_email_textbox.place_forget()
    create_account_password_textbox.place_forget()

```

```

create_account_phno_textbox.place_forget()
log_in_window_label.place(x=0,y=0)
email_id_textbox.place(x=500,y=400)
password_textbox.place(x=500,y=540)
log_in_button.place(x=600,y=625)
create_account_button.place(x=760,y=735)

def log_out():
    is_logout=messagebox.askyesno('News Scoops','Do you want to log out?')
    if is_logout==1:
        destroy_title_bar()
        menu_forget()
        news_frame1.destroy()
        weather_time_frame.destroy()
        time_label.destroy()
        log_in_window_label.place(x=0,y=0)
        email_id_textbox.place(x=500,y=400)
        password_textbox.place(x=500,y=540)
        log_in_button.place(x=600,y=625)
        create_account_button.place(x=760,y=735)

        messagebox.showinfo('News Scoops','You have been logged out')
    else:
        pass

weekday_list=['Monday','Tuesday','Wednesday','Thursday','Friday','Saturday','Sunday']

def clock():
    weekday=datetime.today().weekday()
    day=weekday_list[weekday]
    datetdy=date.today().strftime("%B %d, %Y")
    hour_clock=time.strftime("%I")
    minute_clock=time.strftime("%M")
    second_clock=time.strftime("%S")
    ampm_clock=time.strftime("%p")
    time_label.config( text = f {day} {datetdy}\n\n{int(hour_clock)}:{minute_clock}:{second_clock} {ampm_clock}')
    time_label.after(1000,clock)

def back_button_title_comm():

    global in_which,where_list
    where_list = [g for g,_ in itertools.groupby(where_list)]
    if len(where_list) !=1:
        news_frame1.destroy()
        populate(where_list[len(where_list)-2])
        which_news_label.configure(text=where_list[len(where_list)-2].upper())
        in_which = where_list[len(where_list)-2]
        del where_list[len(where_list)-1]
        if menu_frame_info=='o-i':
            menu_mover()

def create_title_bar():

    global title_bar_main_window_label,menu_button,reload_button,back_button,back_button_title,get_pdf_button
    title_bar_main_window_label=Label(window,image=title_bar_main_window_img)
    menu_button=Button(window,image=menu_button_img,borderwidth=0,bd=0,highlightthickness=0,command=
    menu_mover)
    reload_button=Button(window,image=reload_button_img,borderwidth=0,bd=0,highlightthickness=0,
    command=reload)
    back_button=Button(window,image =back_button_img,borderwidth=0,bd=0,highlightthickness=0,
    command=back_button_cmd)
    back_button_title=Button(window,image = back_button_title_img,borderwidth=0,bd=0,highlightthickness=0,
    command=back_button_title_comm)

```

```

get_pdf_button=Button(window,image=get_pdf_img,borderwidth=0,bd=0,highlightthickness=0,command=get_pdf)
title_bar_main_window_label.pack()
menu_button.place(x=10,y=10)
get_pdf_button.place(x=100,y=10)
back_button_title.place(x=1310,y=10)
reload_button.place(x=1425,y=10)

def destroy_title_bar():

    menu_button.destroy()
    get_pdf_button.destroy()
    back_button.destroy()
    back_button_title.destroy()
    reload_button.destroy()
    title_bar_main_window_label.destroy()

# Creating Loading Window Elements

loading_window_bg = Image.open('pics/News Scoops Loading Screen.png')
resized_loading_window_bg=loading_window_bg.resize((1280,720),Image.ANTIALIAS)
loading_window_final_bg=ImageTk.PhotoImage(image=resized_loading_window_bg)
loading_window_label= Label(window,image=loading_window_final_bg)
loading_window_label.place(x=0,y=0)
loading_bar = ttk.Progressbar( window,orient= HORIZONTAL ,length=600,mode='determinate')
loading_bar.place(x=350,y=550)

def start_loading():

    for i in range(1,101):
        loading_bar['value']=i
        window.update()
        time.sleep(0.07)
    return True

loading_is_over = start_loading()

# Creating Login Screen Elements

def email_id_textbox_info_place(e):
    global email_id_textbox_info
    email_id_textbox_info=Label(window,text='Enter Your Email Id',font=('Verdana',16),background='#242424')
    email_id_textbox_info.place(x=500,y=375,height=25)

def password_textbox_info_place(e):
    global password_textbox_info
    password_textbox_info=Label(window,text='Enter Your Password',font=('Verdana',16),background='#242424')
    password_textbox_info.place(x=500,y=515,height=25)

log_in_window_bg = Image.open('pics\Log In bg.png')
resized_log_in_window_bg=log_in_window_bg.resize((w,h),Image.ANTIALIAS)
log_in_window_final_bg=ImageTk.PhotoImage(image=resized_log_in_window_bg)
log_in_window_label=Label(window,image=log_in_window_final_bg,borderwidth=0)
email_id_textbox=ttk.Entry(window,width=28,font=('Verdana',25))
email_id_textbox.bind('<Enter>',email_id_textbox_info_place)
email_id_textbox.bind('<Leave>',lambda e: email_id_textbox_info.destroy())
password_textbox=ttk.Entry(window,width=28,font=('Verdana',25),show=chr(0x2022))
password_textbox.bind('<Enter>',password_textbox_info_place)
password_textbox.bind('<Leave>',lambda e: password_textbox_info.destroy())
log_in_button_bg=ImageTk.PhotoImage(file='pics/LOGIN BUTTON.png')
log_in_button=Button(window,image=log_in_button_bg,borderwidth=0,bd=0,highlightthickness=0,relief=FLAT,background='#212121',command=log_in_main_window)
create_account_button_bg=ImageTk.PhotoImage(file='pics/Create Account Button.png')
create_account_button=Button(window,image=create_account_button_bg,borderwidth=0,bd=0,relief=FLAT,

```

```

highlightthickness=0,command=create_account_navigation)

if loading_is_over:
    def do_stuff():
        window.overrideredirect(False)
        window.geometry(f'{w}{x}{h}+0+0')
        loading_window_label.destroy()
        loading_bar.destroy()
        log_in_window_label.place(x=0,y=0)
        email_id_textbox.place(x=500,y=400)
        password_textbox.place(x=500,y=540)
        log_in_button.place(x=600,y=625)
        create_account_button.place(x=760,y=735)
    window.after(3000,do_stuff)

# NEWS WINDOW #
#Menu Images

top_headlines_button_img=ImageTk.PhotoImage(file='pics\TOP HEADLINES BUTTON.png')
business_button_img=ImageTk.PhotoImage(file='pics\BUSINESS BUTTON.png')
entertainment_button_img=ImageTk.PhotoImage(file='pics\ENTERTAINMENT BUTTON.png')
science_button_img=ImageTk.PhotoImage(file='pics\SCIENCE BUTTON.png')
technology_button_img=ImageTk.PhotoImage(file='pics\TECHNOLOGY BUTTON.png')
sport_button_img=ImageTk.PhotoImage(file='pics\SPORTS BUTTON.png')
health_button_img=ImageTk.PhotoImage(file='pics\HEALTH BUTTON.png')
log_out_button_img=ImageTk.PhotoImage(file='pics\LOG OUT BUTTON.png')

# Title Bar
title_bar_main_window_img=ImageTk.PhotoImage(file='pics\TITLE BAR LABEL.png')
menu_button_img=ImageTk.PhotoImage(file='pics\MENU BUTTON.png')
reload_button_img=ImageTk.PhotoImage(file='pics\RELOAD.png')
back_button_img=ImageTk.PhotoImage(file='pics\BACK BUTTON.png')
back_button_title_img=ImageTk.PhotoImage(file='pics\Back Button Title.png')
get_pdf_img=ImageTk.PhotoImage(file='pics\GET PDF.png')

# Create Account Window

def create_account_name_info_place(e):
    global create_account_name_info
    create_account_name_info=Label(window,text='Enter Your Name',font=('Verdana',16),background='#242424')
    create_account_name_info.place(x=550,y=320,height=25)
def create_account_email_info_place(e):
    global create_account_email_info
    create_account_email_info=Label(window,text='Enter Your Email',font=('Verdana',16),background='#242424')
    create_account_email_info.place(x=550,y=410,height=25)
def create_account_password_info_place(e):
    global create_account_password_info
    create_account_password_info=Label(window,text='Enter a Strong Password within 20 characters',
    font=('Verdana',16),background='#242424')
    create_account_password_info.place(x=550,y=500,height=25)
def create_account_phno_info_place(e):
    global create_account_phno_info
    create_account_phno_info=Label(window,text='Enter Your 10-digit Phone    Number',font=('Verdana',16)
    ,background='#242424')
    create_account_phno_info.place(x=550,y=590,height=25)

create_account_window_img=ImageTk.PhotoImage(file='pics\CREATE ACCOUNT BG.png')
create_account_window_label= Label(window,image=create_account_window_img,
borderwidth=0,bd=0,highlightthickness=0)
create_account_window_create_account_button_img=ImageTk.PhotoImage(file='pics\CREATE ACCOUNT BUTTON
CREATE ACCOUNT WINDOW.png')
create_account_window_create_account_button=Button(window,image=create_account_window_create_account_button
_img,borderwidth=0,bd=0,highlightthickness=0,command=create_account_window_create_account)

```

```

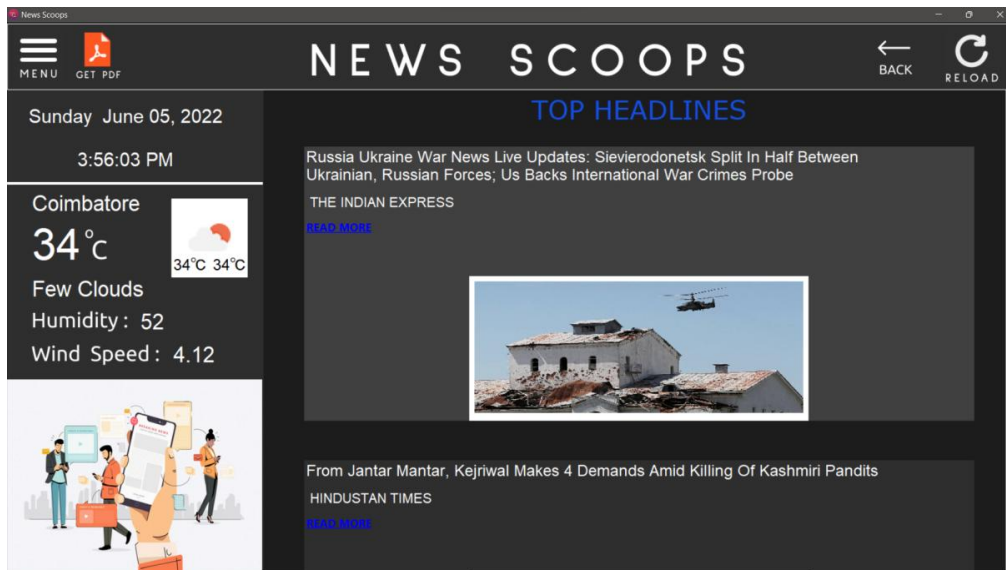
create_account_back_button_img=ImageTk.PhotoImage(file='pics\BACK BUTTON 2.png')
create_account_back_button=Button(window,image=create_account_back_button_img,borderwidth=0,bd=0,
highlightthickness=0,command=create_account_back)
create_account_name_textbox=ttk.Entry(window,width=38,font=('Verdana',25))
create_account_name_textbox.bind('<Enter>',create_account_name_info_place)
create_account_name_textbox.bind('<Leave>',lambda e:create_account_name_info.destroy())
create_account_email_textbox=ttk.Entry(window,width=38,font=('Verdana',25))
create_account_email_textbox.bind('<Enter>',create_account_email_info_place)
create_account_email_textbox.bind('<Leave>',lambda e:create_account_email_info.destroy())
create_account_password_textbox=ttk.Entry(window,width=38,font=('Verdana',25),show=chr(0x2022))
create_account_password_textbox.bind('<Enter>',create_account_password_info_place)
create_account_password_textbox.bind('<Leave>',lambda e: create_account_password_info.destroy())
create_account_phno_textbox=ttk.Entry(window,width=38,font=('Verdana',25))
create_account_phno_textbox.bind('<Enter>',create_account_phno_info_place)
create_account_phno_textbox.bind('<Leave>',lambda e: create_account_phno_info.destroy())
window.mainloop()

```

OUTPUT

OUTPUT

MAIN WINDOW



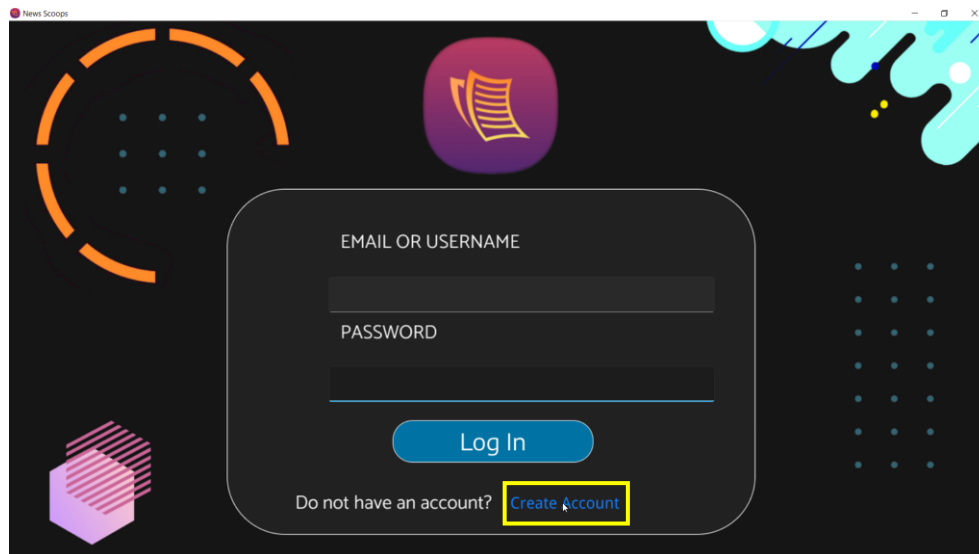
This is the main window of the **NEWS SCOOPS** application. It shows **TOP HEADLINES** as the default news.

SPLASH SCREEN



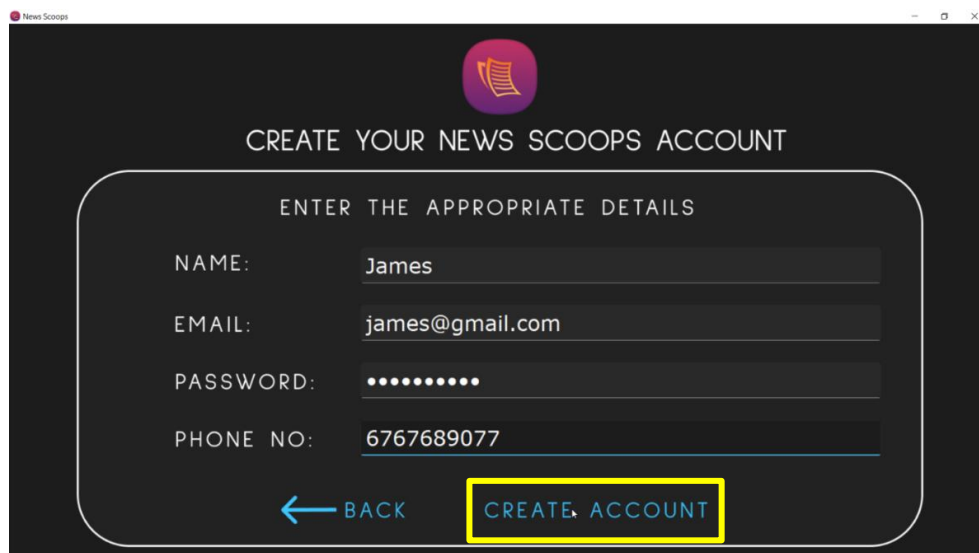
The **NEWS APPLICATION** starts with a SplashScreen, with a quote about news.

LOG IN WINDOW - CREATING ACCOUNT



In the Log in window, a user can enter the correct username and password to login. If the entered information is wrong, then a message box showing a warning will be produced. You can click the create account button to create an account. Clicking the **Create Account Button**, it will direct to the **Create Account Window**.

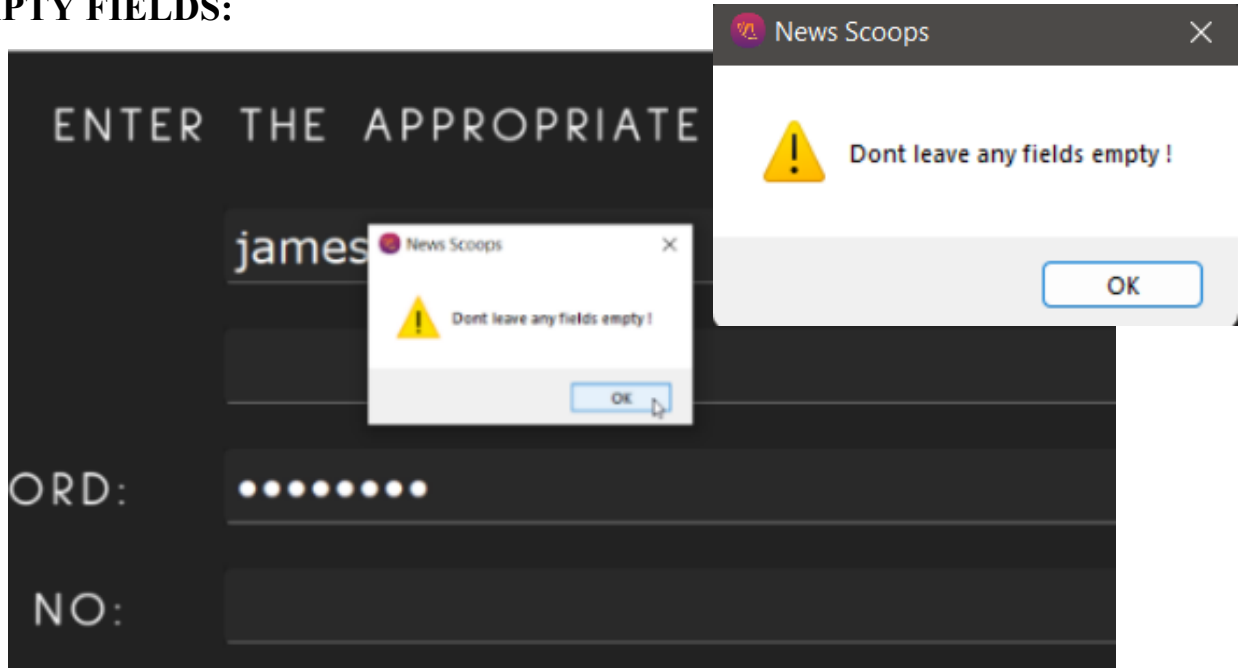
CREATE ACCOUNT WINDOW



In create account window the user can enter the required details and create a new **NEWS SCOOPS** account.

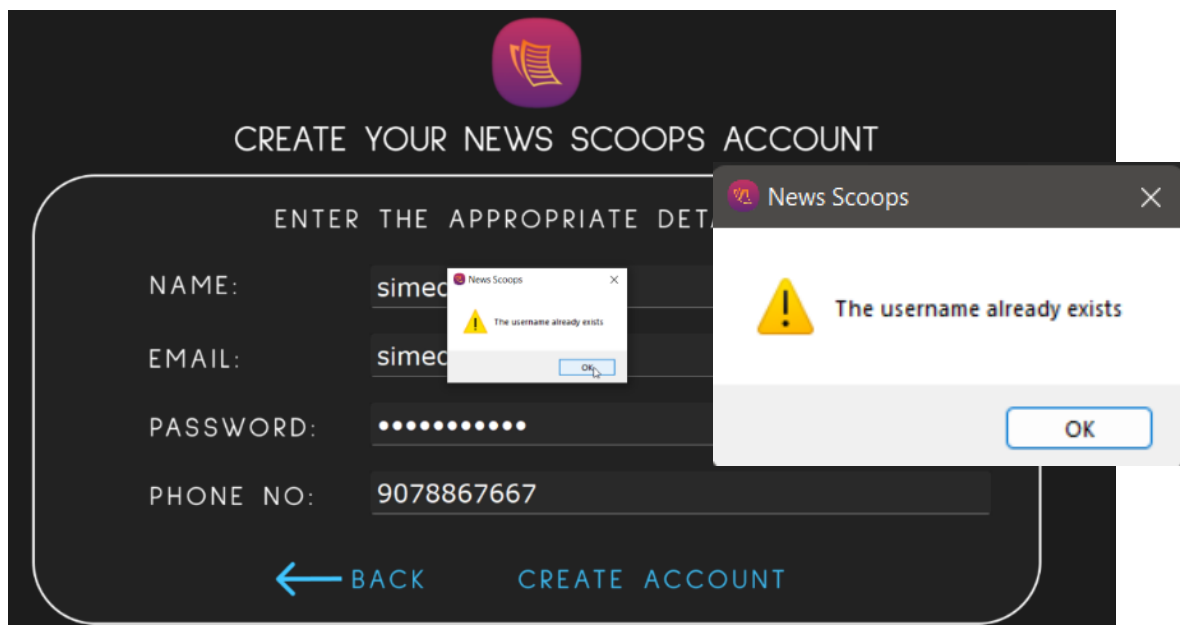
CREATE ACCOUNT WINDOW – (Wrong Credentials)

1. EMPTY FIELDS:



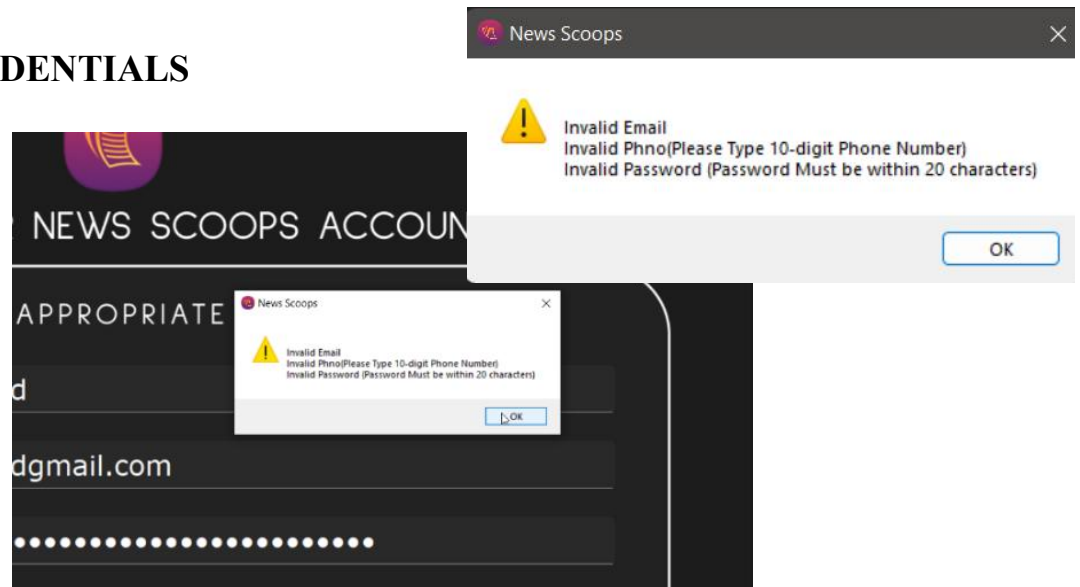
If the fields are empty the application will show a warning **‘THE FIELDS ARE EMPTY!’**.

2.USERNAME (EMAIL) ALREADY EXISTS



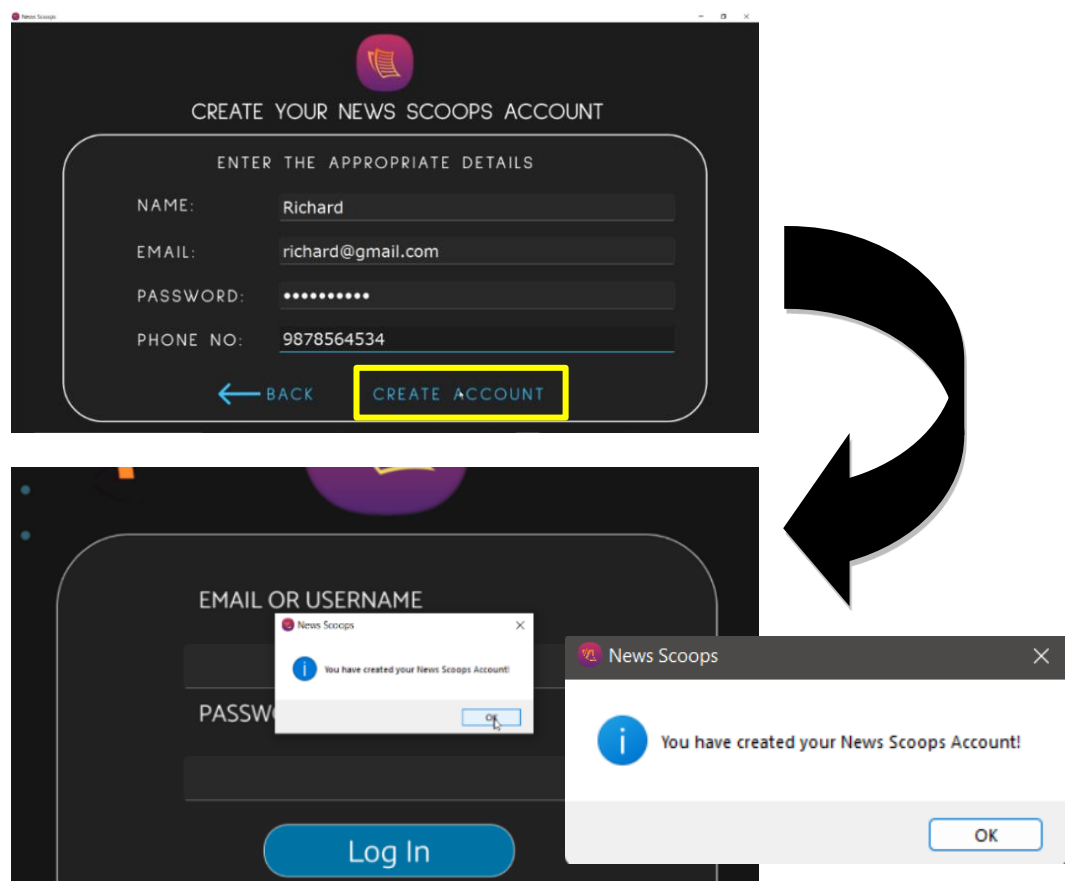
If the username which is the email address of the user already exists the application will show a warning **‘THE USERNAME ALREADY EXISTS!’**.

3.WRONG CREDENTIALS



If the wrong credentials are given then the application will show a warning which says which details are wrong.

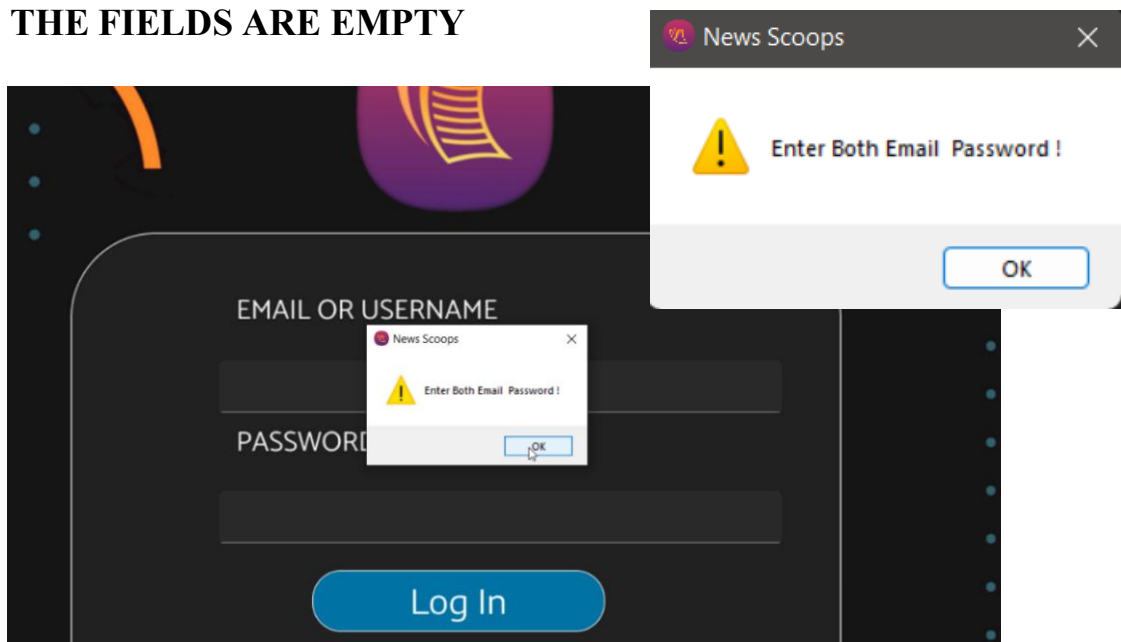
CREATE ACCOUNT - (Correct Credentials)



If the correct credentials are given and if you click the create account button it will direct you to the Log In window with a message '**YOU HAVE CREATED YOUR NEWS SCOOPS ACCOUNT**'.

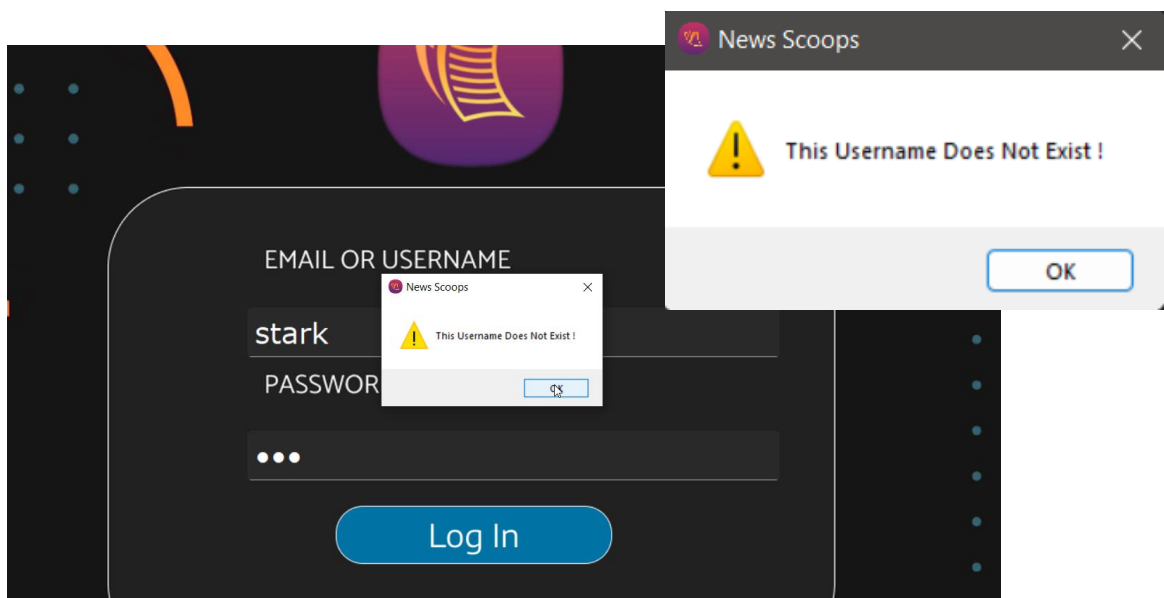
LOGIN WINDOW - (Wrong Credentials)

1.WHEN THE FIELDS ARE EMPTY



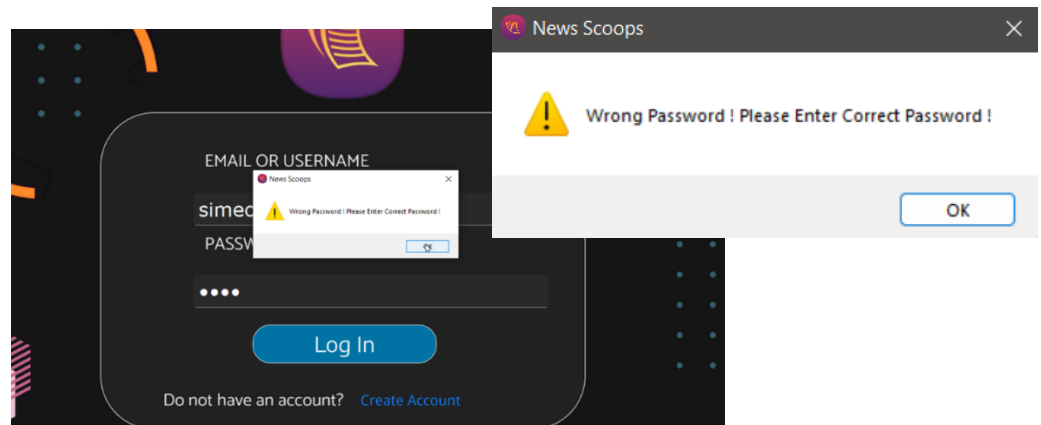
If the fields are empty, it shows a warning **‘ENTER BOTH EMAIL AND PASSWORD!’**.

2.WHEN THE USERNAME DOES NOT EXIST



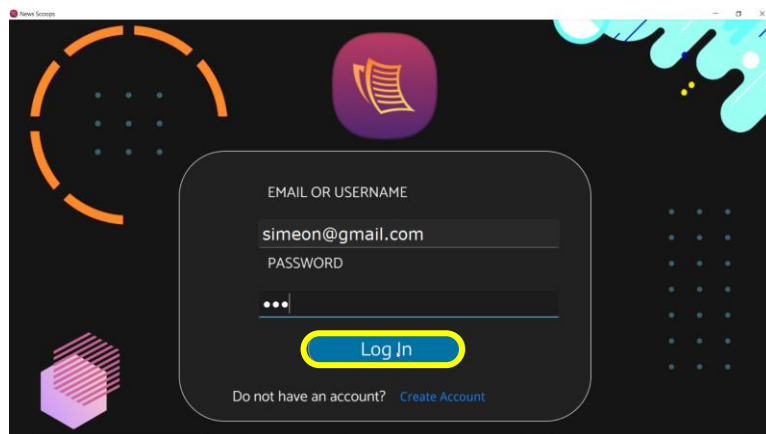
If the user username does not exist, shows a warning **‘THE USERNAME DOES NOT EXIST!’**.

3.WHEN WRONG PASSWORD IS GIVEN



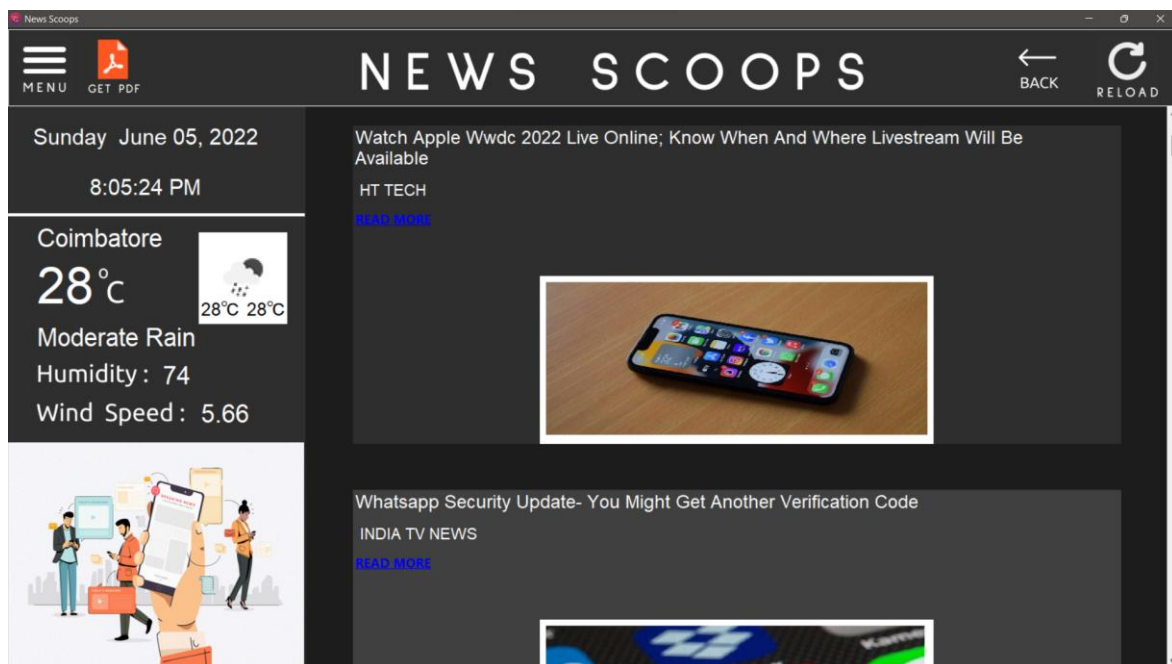
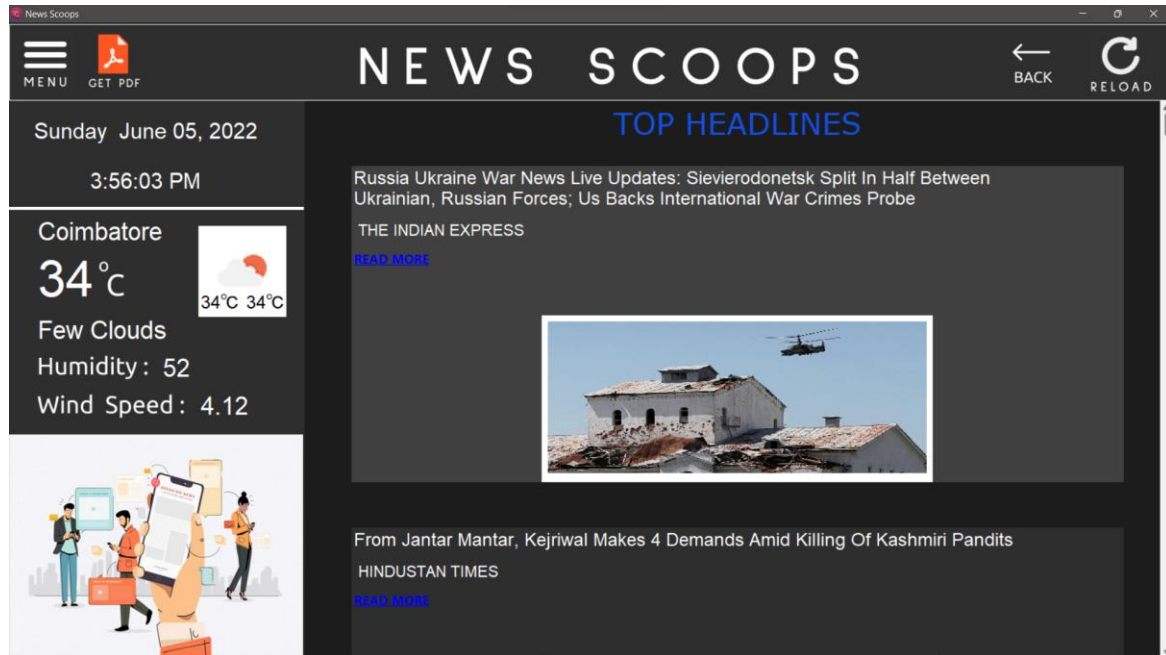
When the Wrong Password is given, it shows a warning **‘WRONG PASSWORD! PLEASE ENTER THE CORRECT PASSWORD!’**

LOGGING IN - (Correct Credentials)



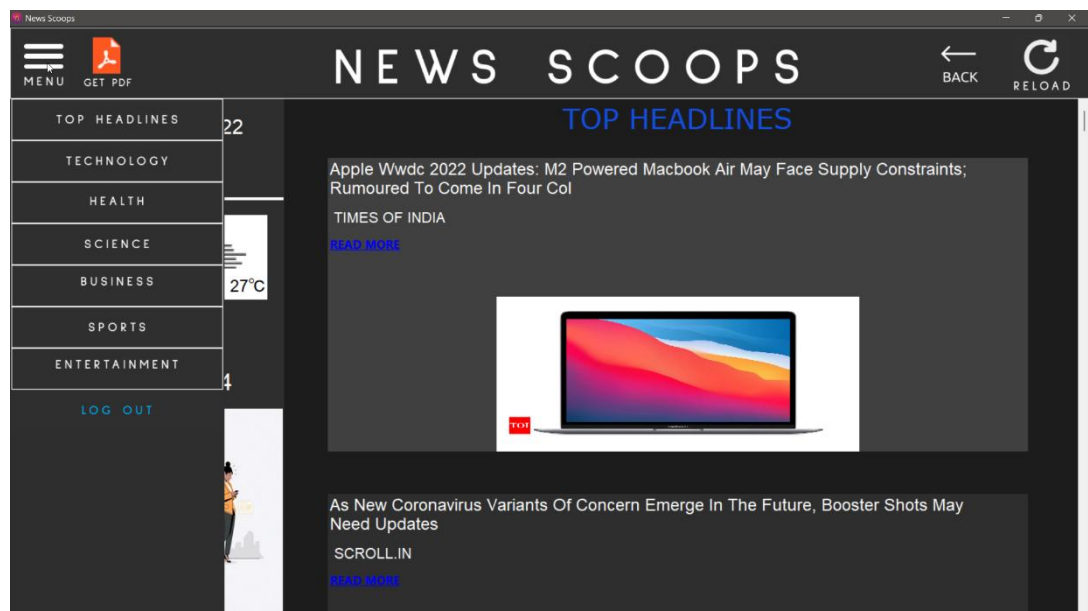
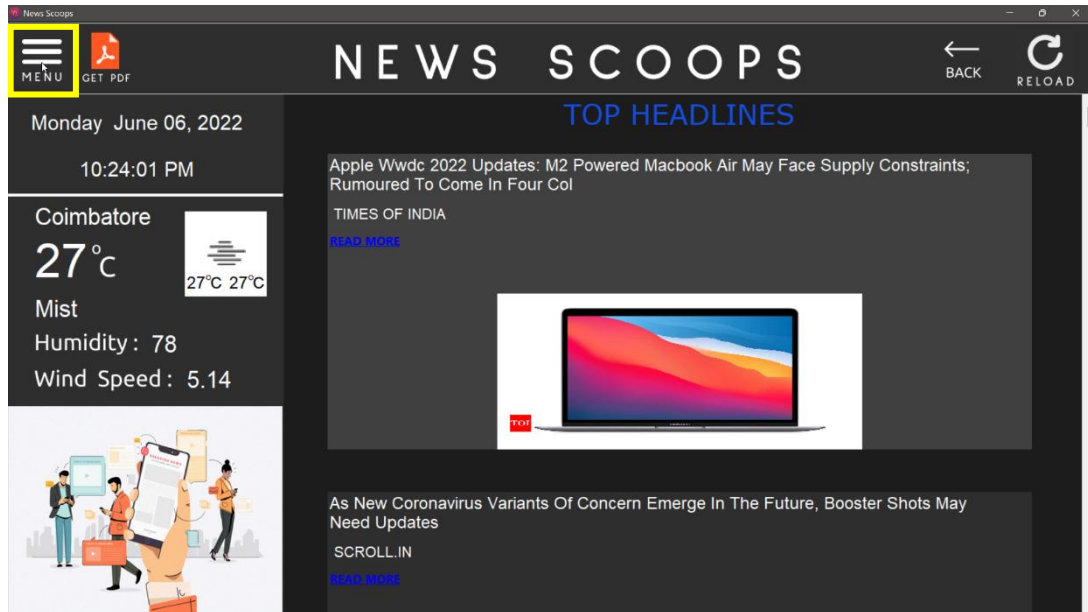
Typing the correct credentials and clicking the Log in Button it directs to the main window.

HOME WINDOW

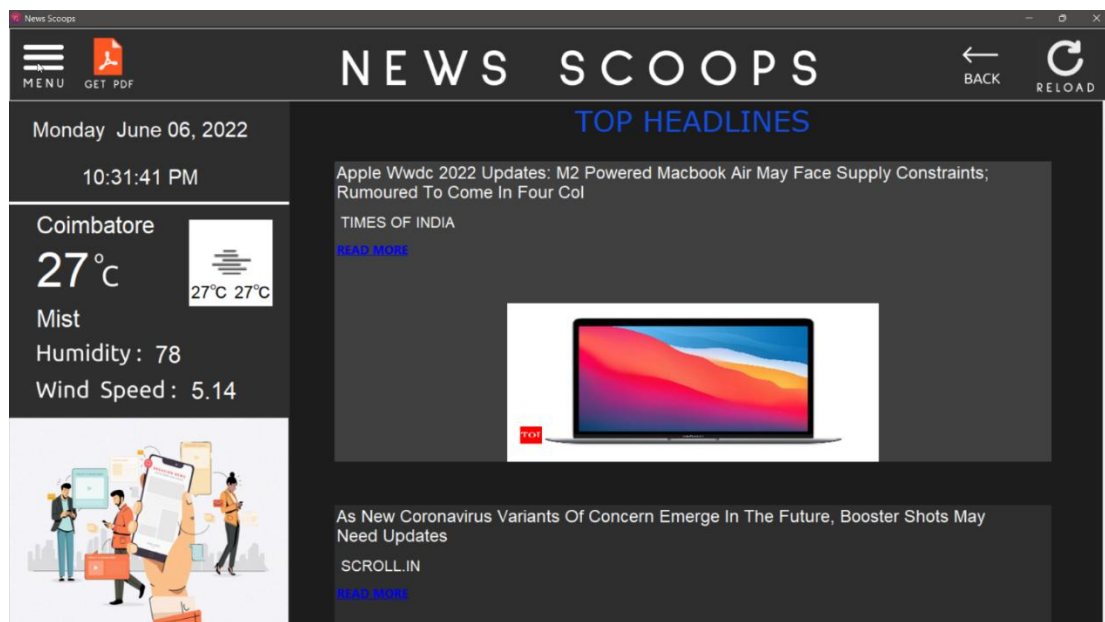
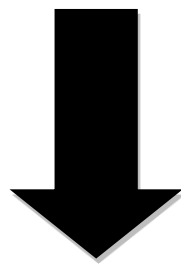
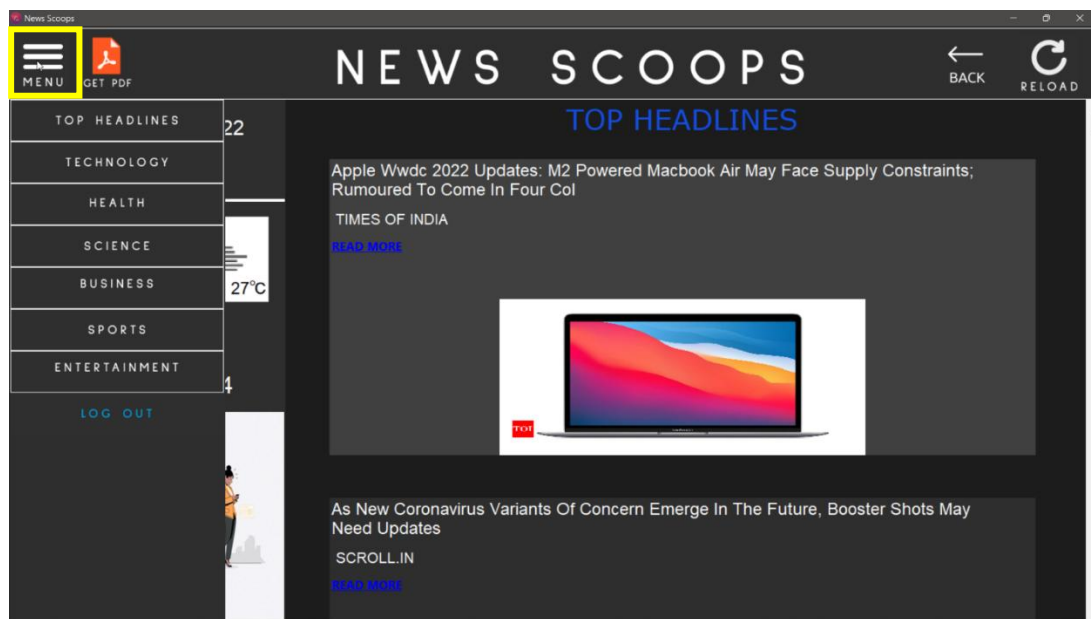


This window is displayed when the user Logs in successfully. This window shows **TOP HEADLINES** as default news.

MENU BUTTON



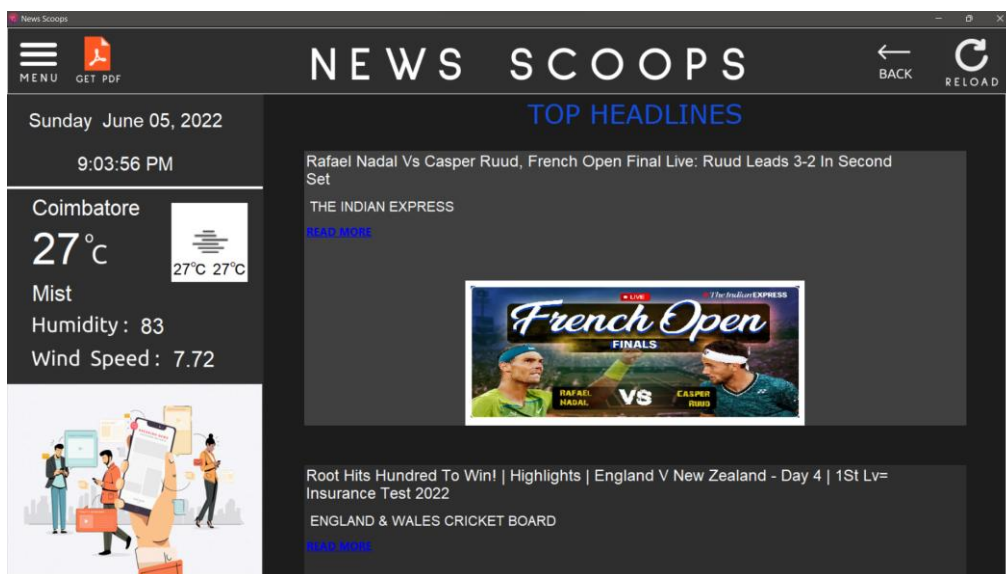
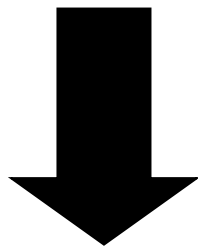
When you click the menu button once the menu will appear by swishing in the window.



When you click the menu button again the menu will swish back and disappear.

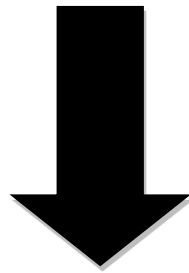
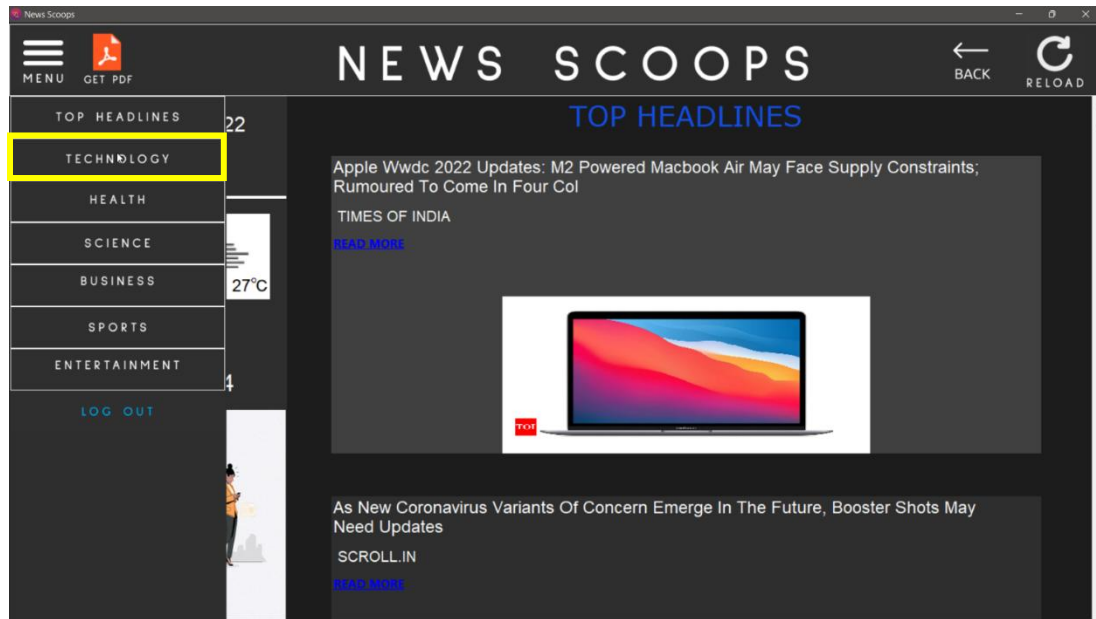
CATEGORY OF NEWS YOU CAN VIEW

1. MENU: TOP HEADLINES - (DEFAULT)



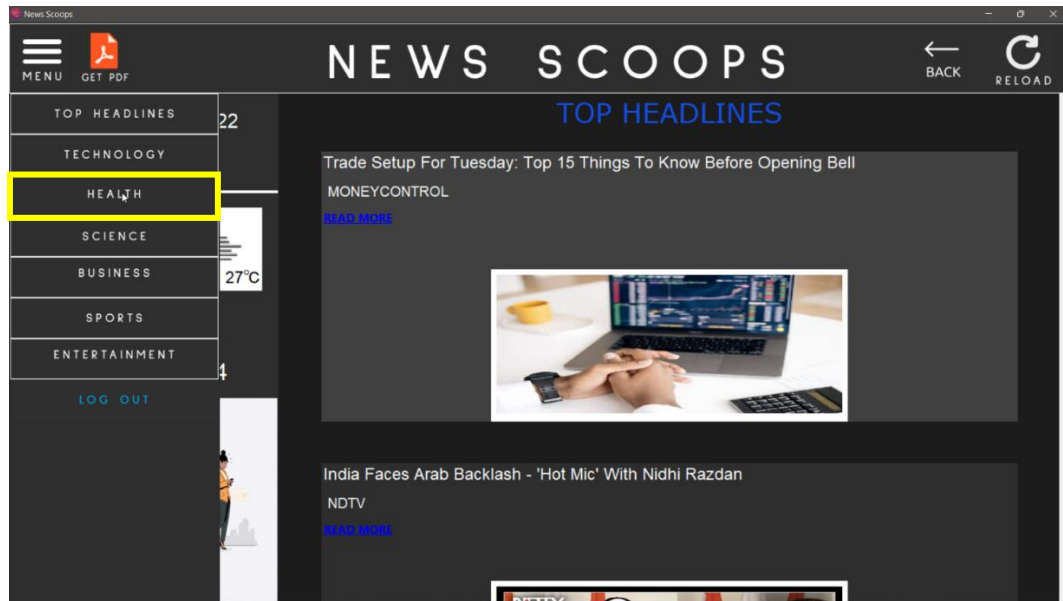
The **TOP HEADLINES** can be viewed by clicking the top headlines button in the menu. The **TOP HEADLINES** will also be shown when you Log In as default.

2. MENU: TECHNOLOGY



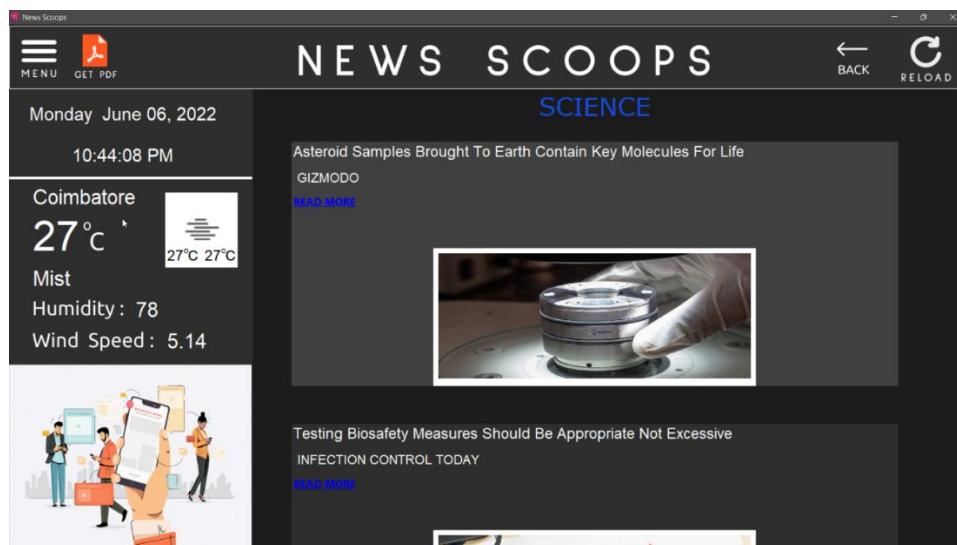
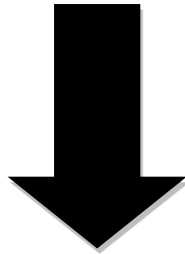
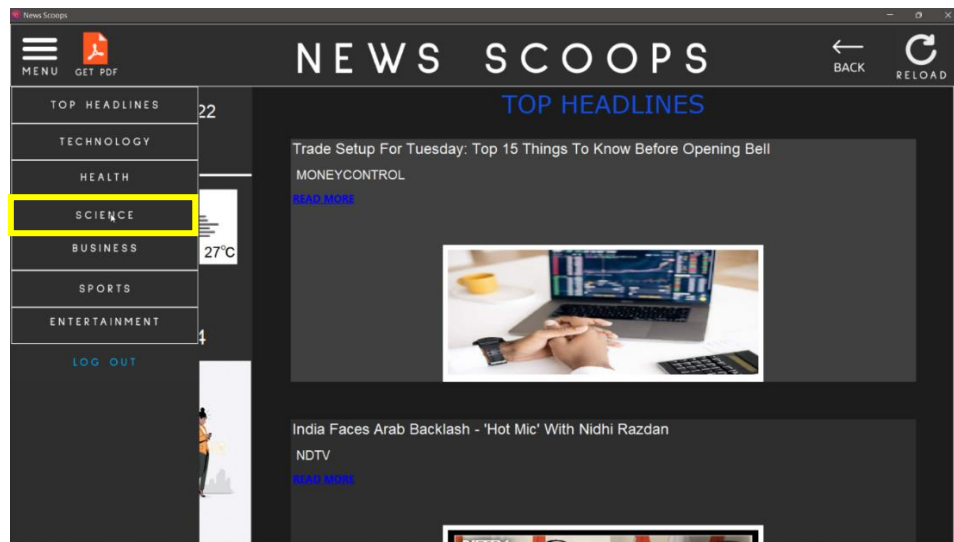
You can view News related to **TECHNOLOGY** by clicking on the **TECHNOLOGY** **BUTTON** in the **MENU**.

3. MENU: HEALTH



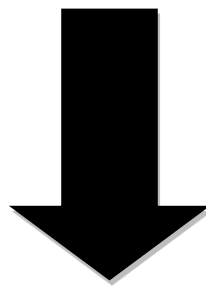
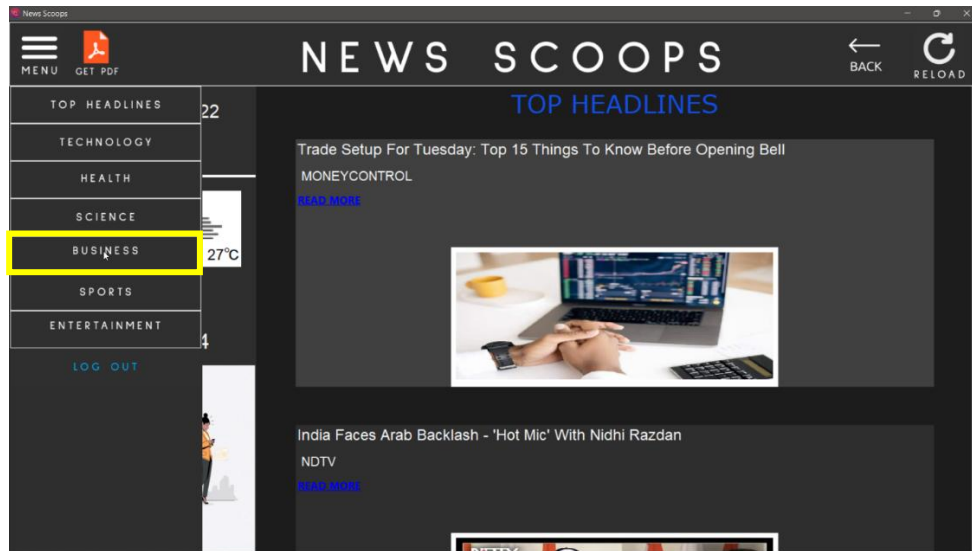
You can view News related to **HEALTH** by clicking on the **HEALTH BUTTON** in the **MENU**.

4. MENU: SCIENCE



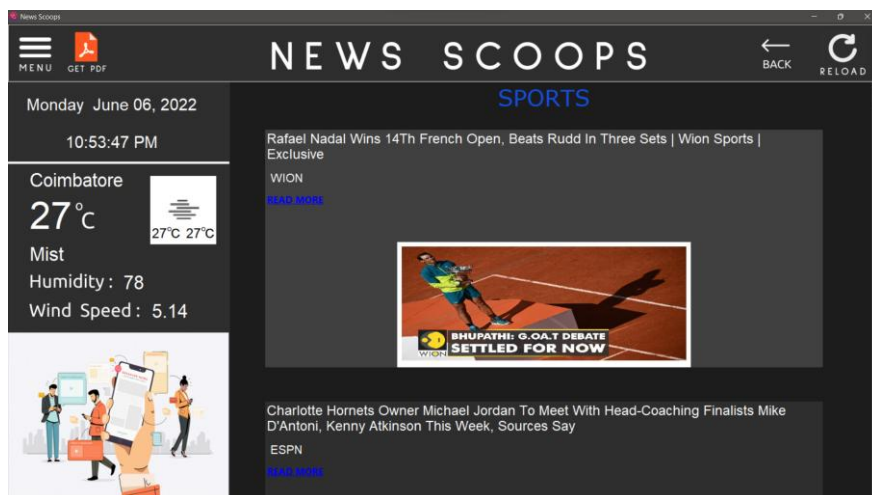
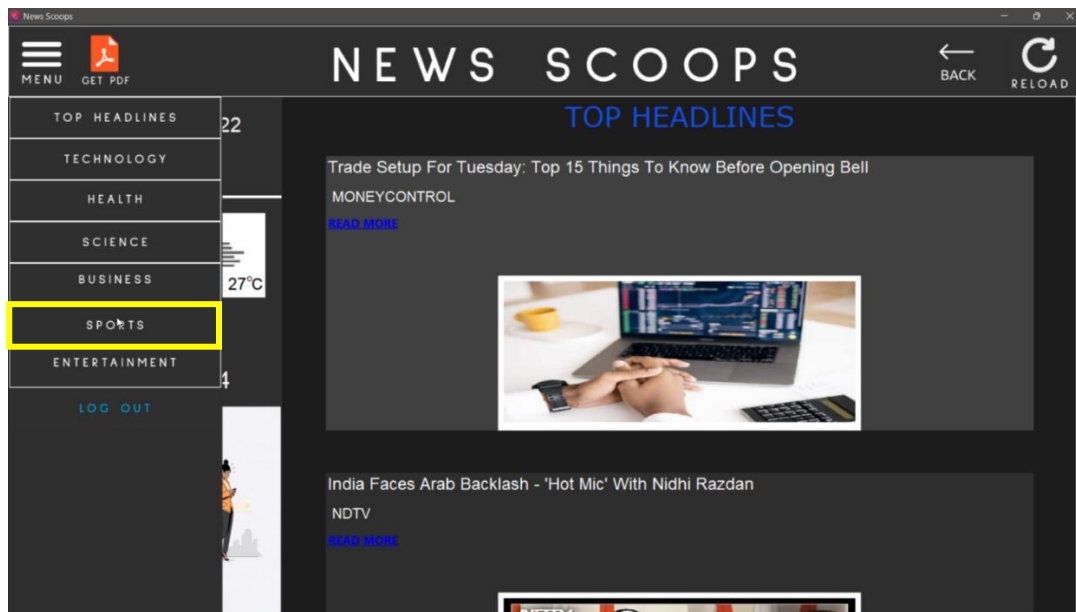
You can view News related to **SCIENCE** by clicking on the **SCIENCE BUTTON** in the **MENU**.

5. MENU: BUSINESS



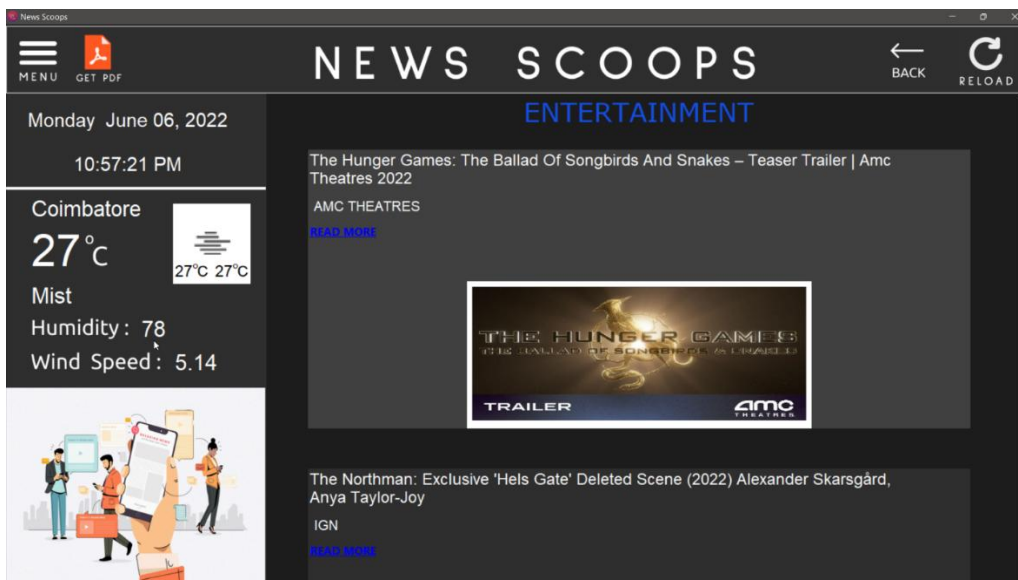
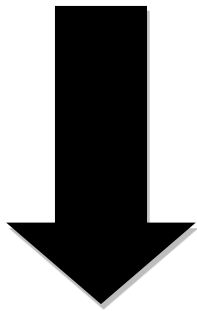
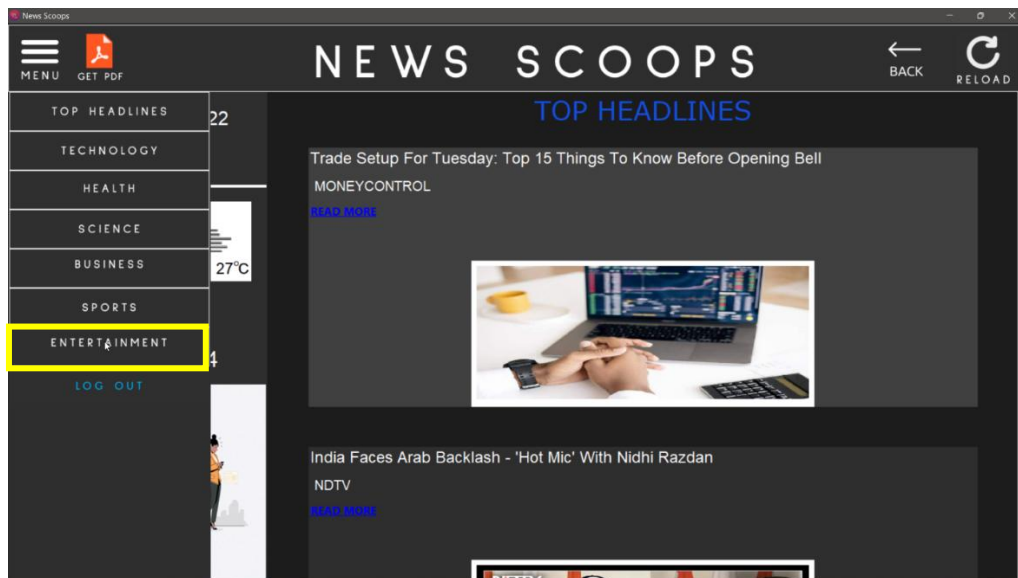
You can view News related to **BUSINESS** by clicking on the **BUSINESS BUTTON** in the **MENU**.

6. MENU: SPORTS



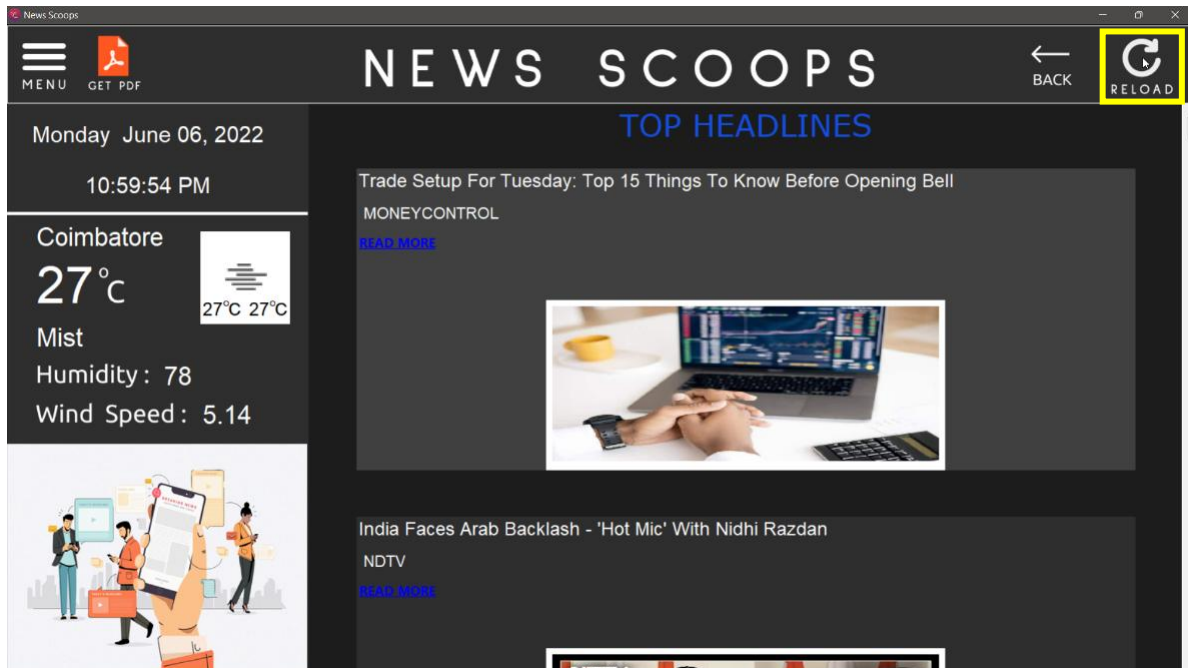
You can view News related to **SPORTS** by clicking on the **SPORTS BUTTON** in the **MENU**.

7. MENU: ENTERTAINMENT



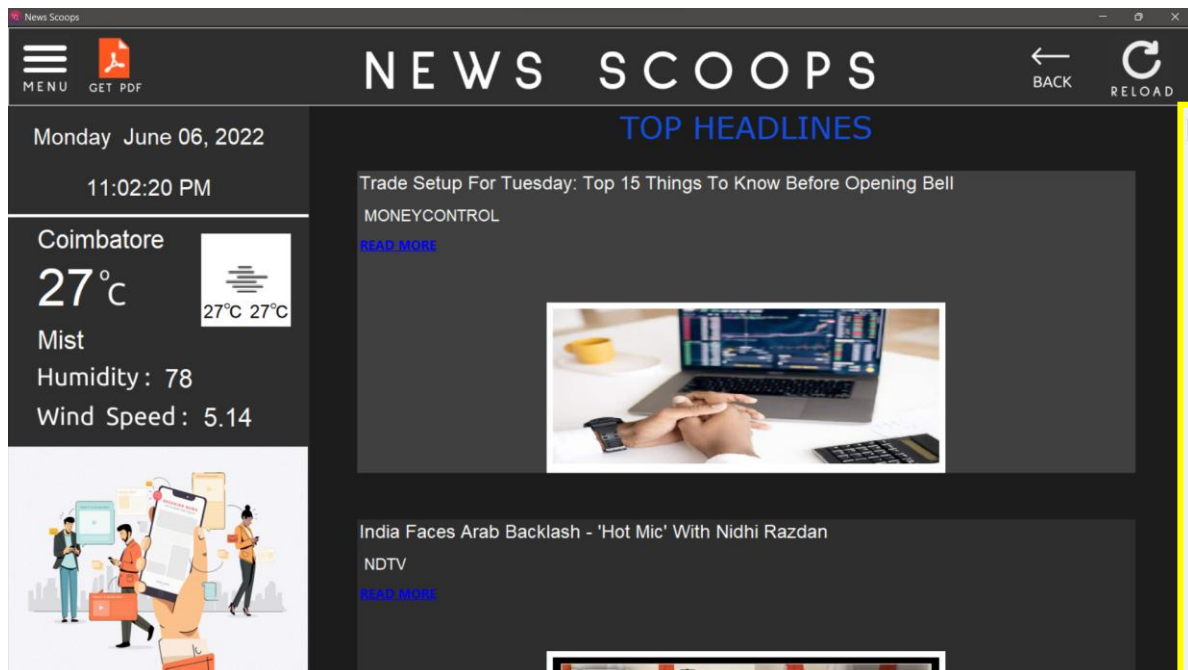
You can view News related to **ENTERTAINMENT** by clicking on the **ENTERTAINMENT BUTTON** in the **MENU**.

RELOADING THE NEWS



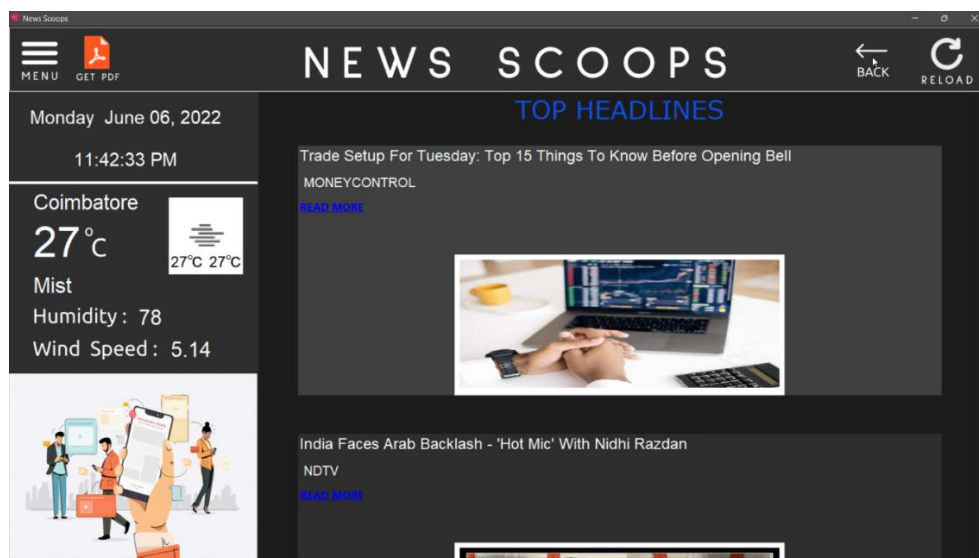
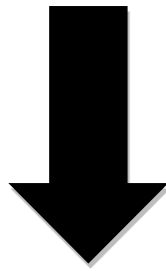
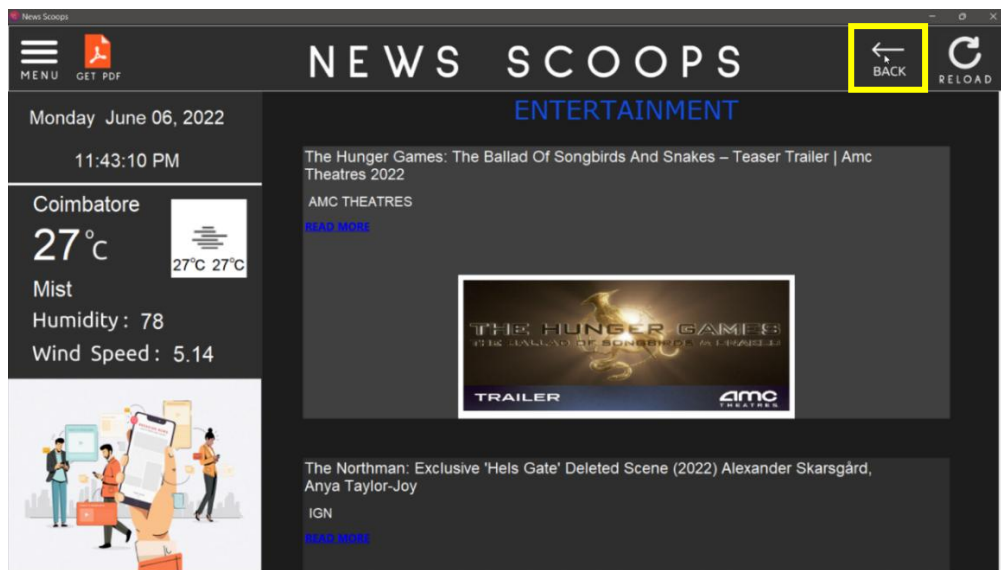
You can Reload the News By clicking the Reload Button.This will refresh the page which you are viewing.

VIEWING THE NEWS



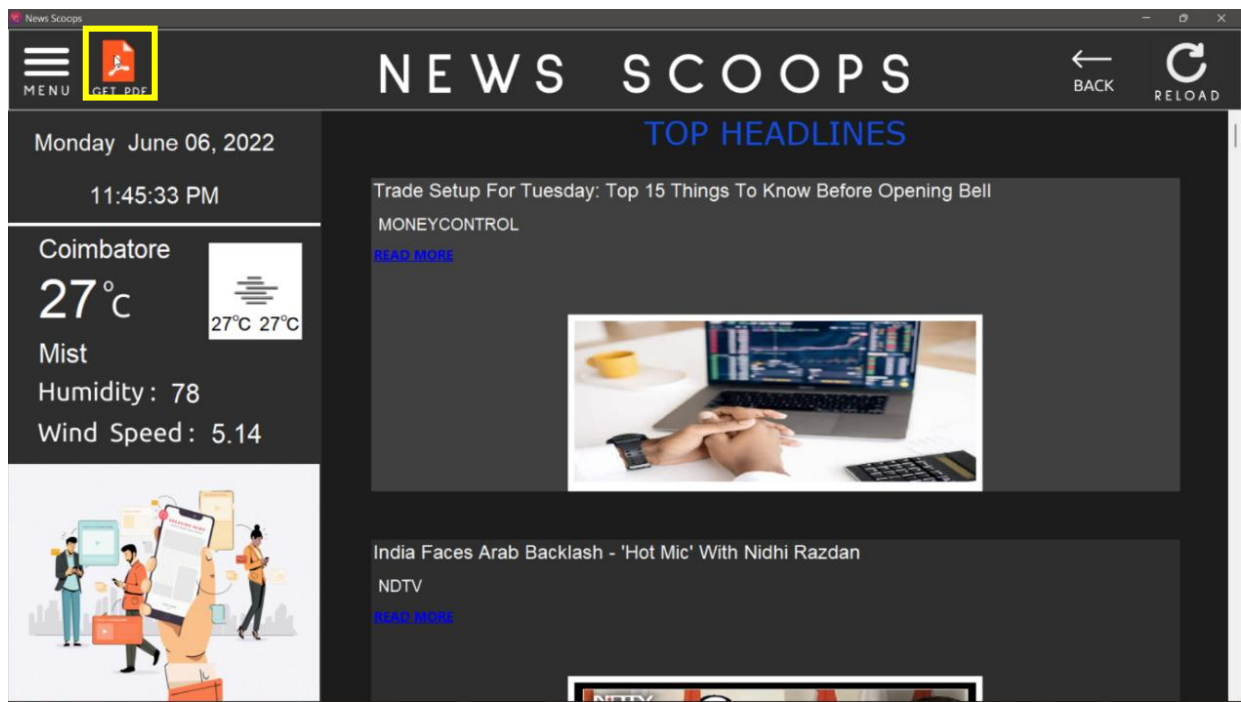
You can view the news by scrolling through the page.

BACK BUTTON

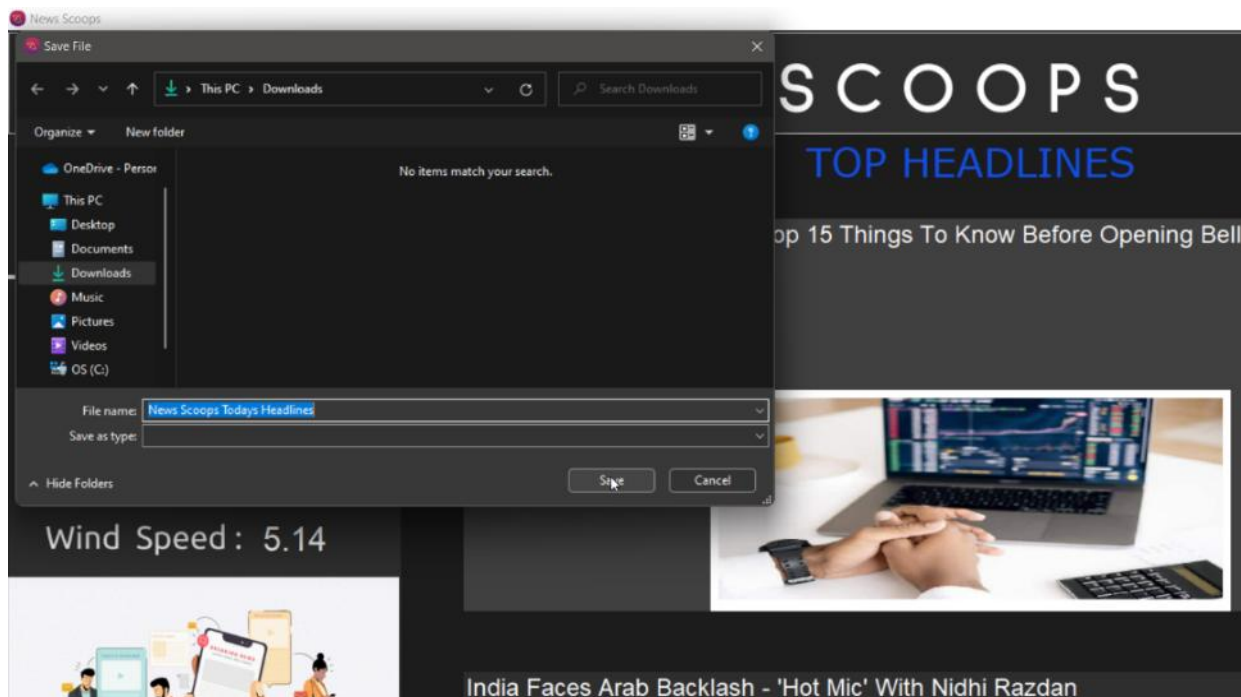


If you click the back button it will take you to the previous news page you had viewed.

GET PDF BUTTON

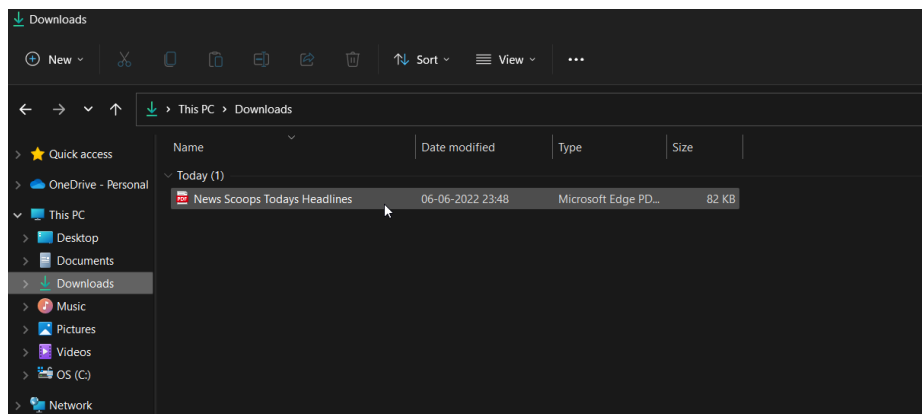


The get pdf button will create a pdf of all the top headlines at that moment.



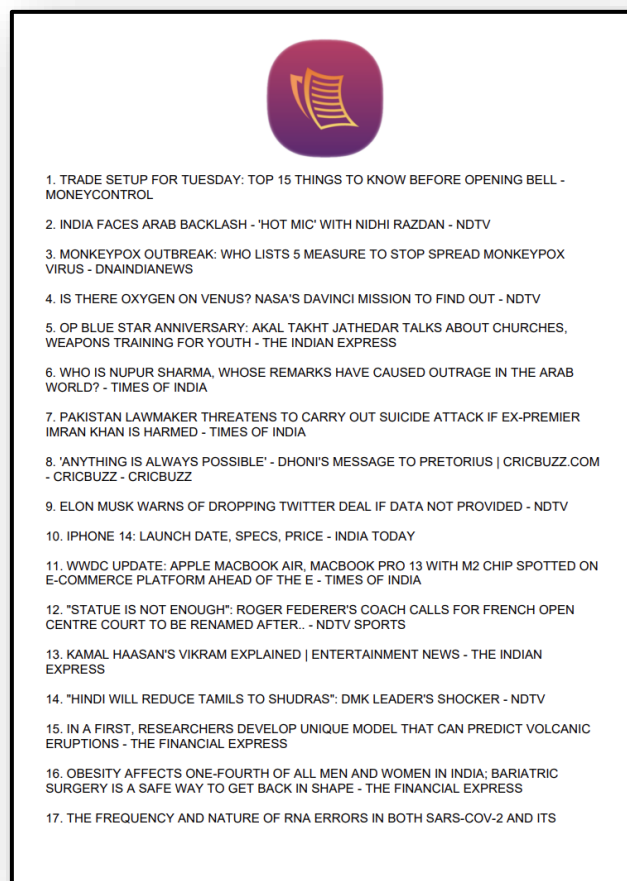
A Save As Dialog Box will appear, select the directory in which you want to save the file and if necessary, type the name of the file.

OPENING PDF FILE



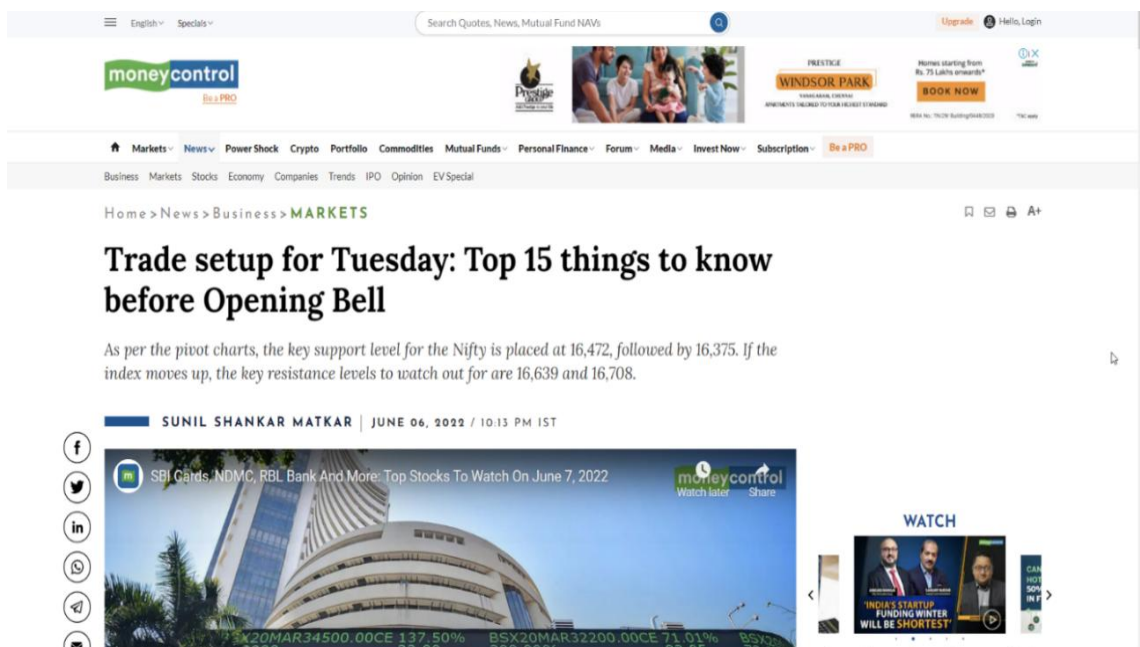
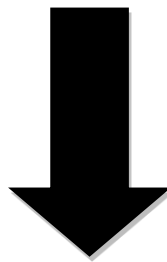
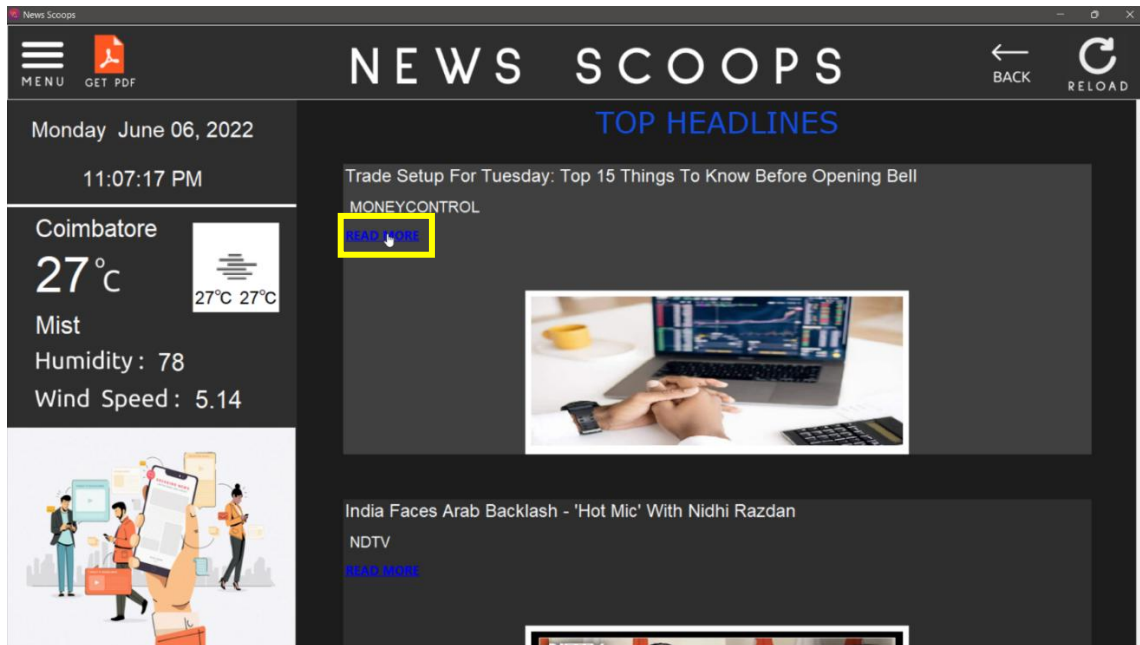
Open the file by double clicking it and you will be able to view the pdf file.

OUTPUT



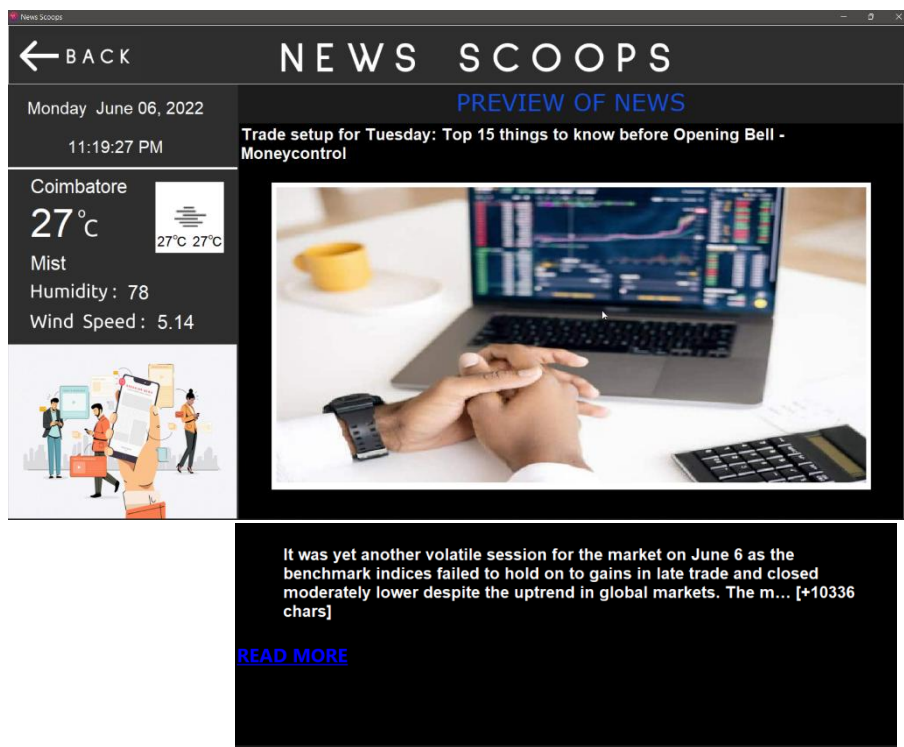
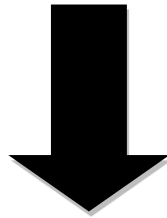
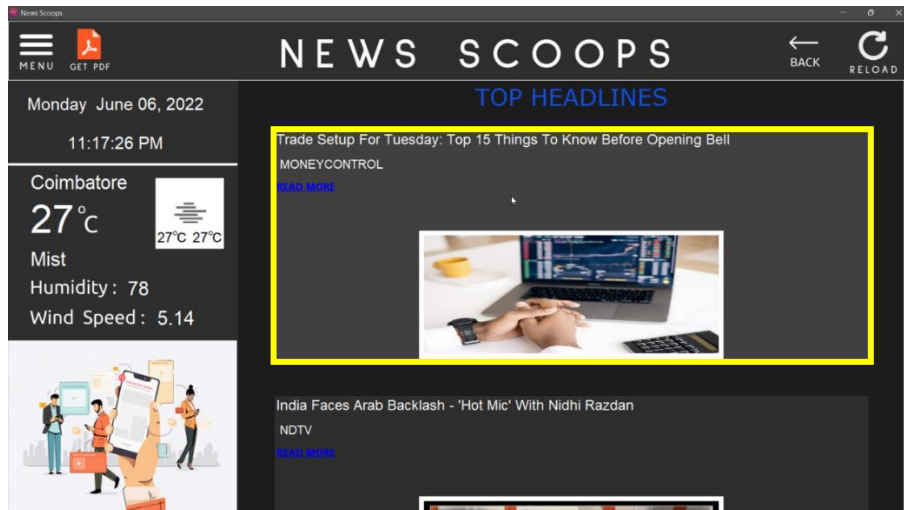
This PDF file will contain top headlines of the current news at that moment when it was created.

CLICKING ON READ MORE



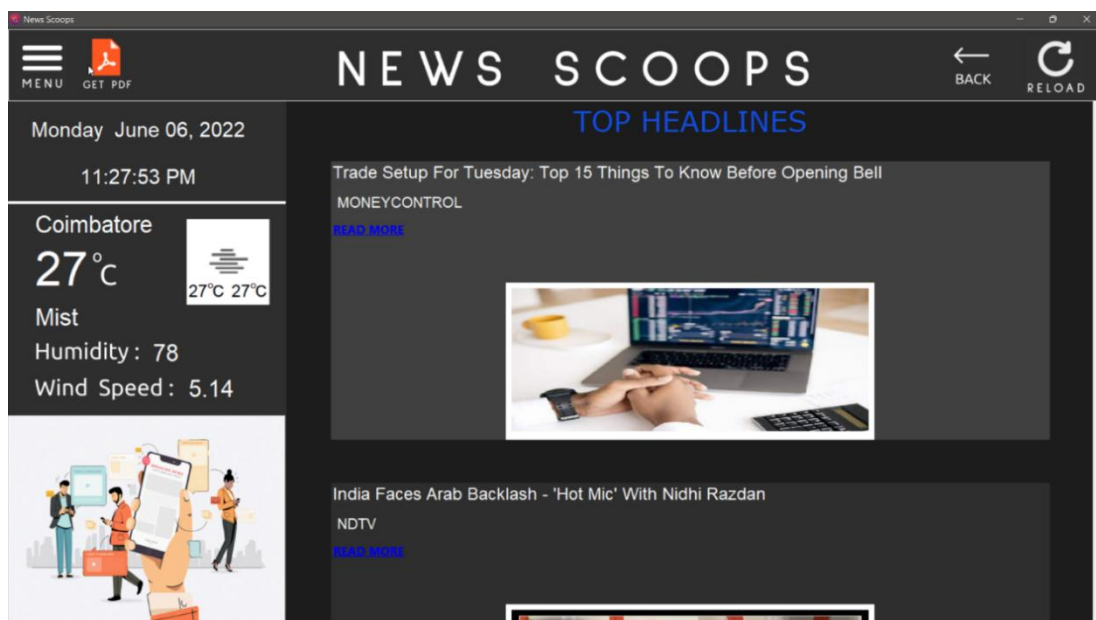
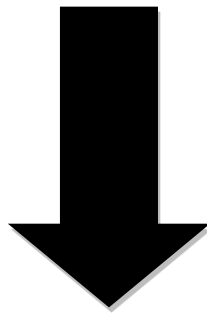
By clicking on read more, this will open your browser and it will redirect to the news source from where the news is taken.

GETTING PREVIEW OF THE NEWS



You can view the preview of the news by clicking on the news. It will redirect you to a page which will give the preview of the news. It has the headlines, image, description and read more button.

BACK BUTTON

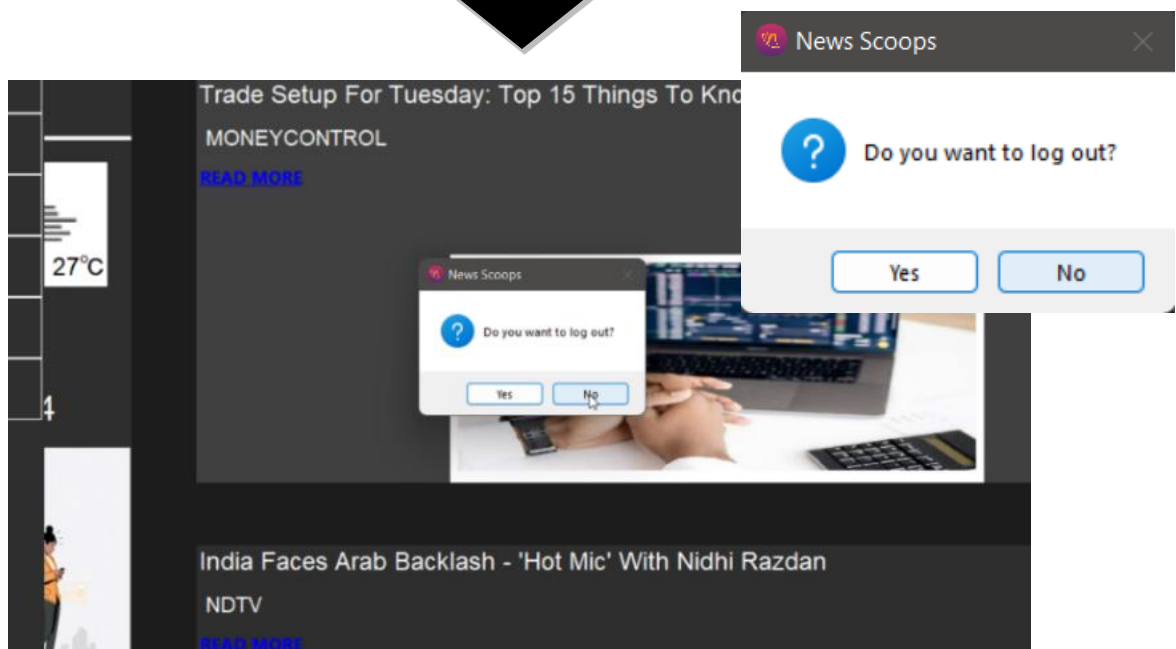
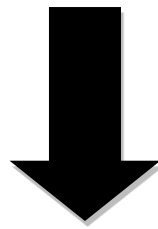
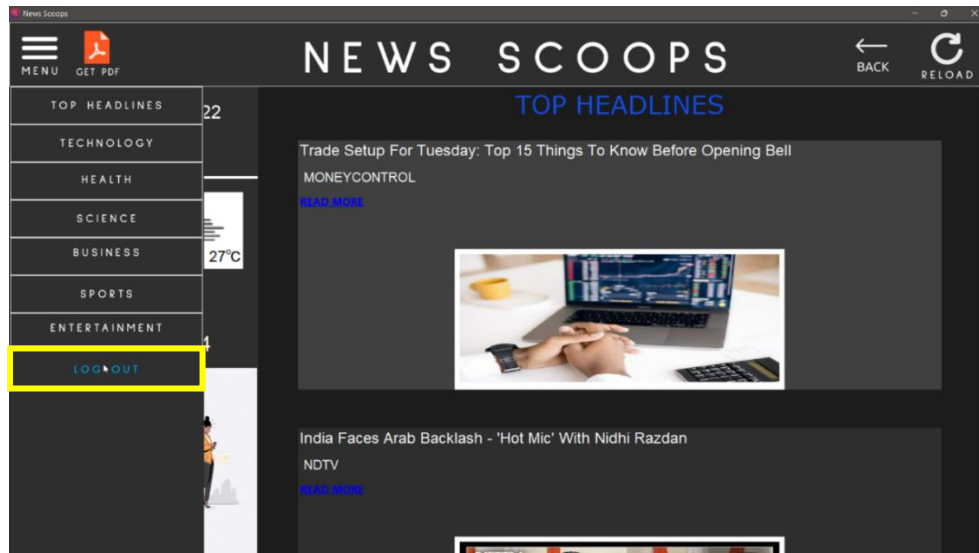


By clicking on the back button, it will redirect you back to the page where you can see the news .

LOG OUT

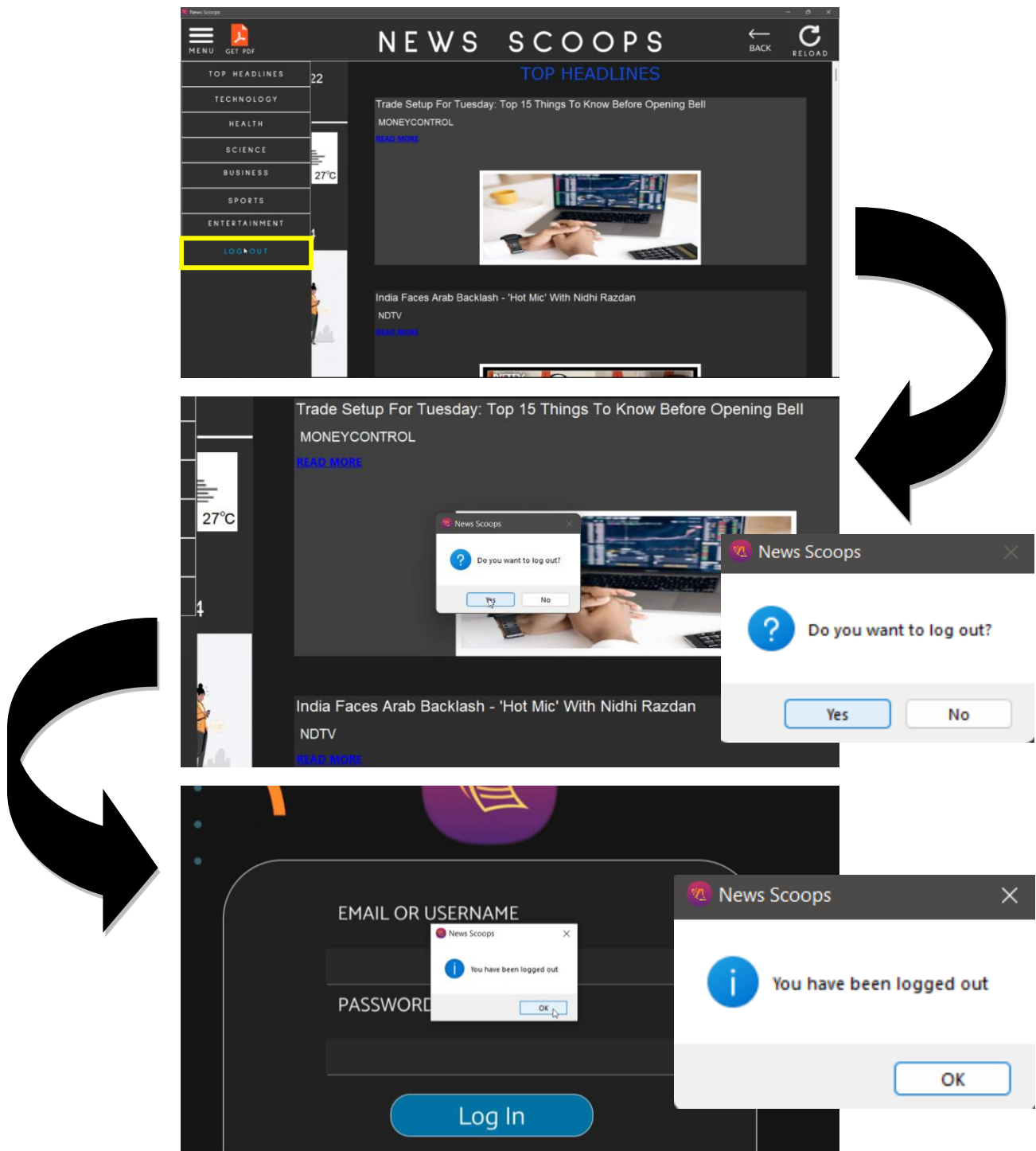
You can log out by clicking the log out button in the menu. It will show a dialogue box asking ‘**DO YOU WANT TO LOG OUT ?**’, with options yes or no.

1. LOG OUT: CLICKING NO



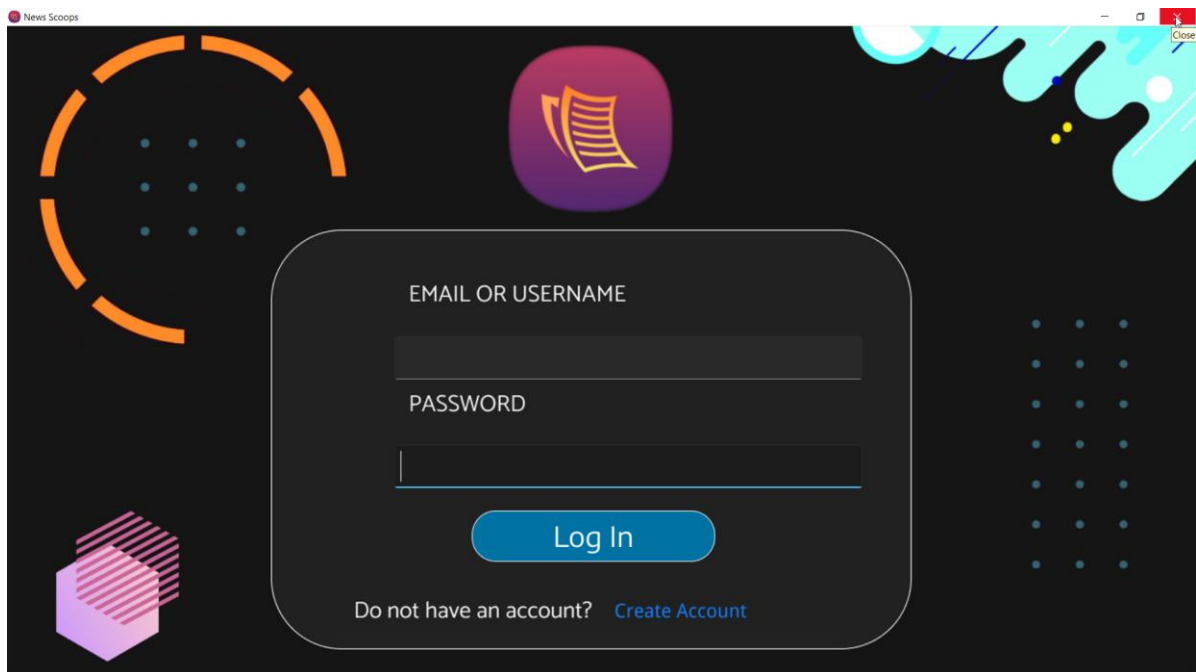
By clicking no you will not be logged out.

2. LOG OUT : CLICKING YES



By clicking yes you will be logged out and you will be redirected to the login window with a message 'You have been logged out'.

CLOSING THE APPLICATION



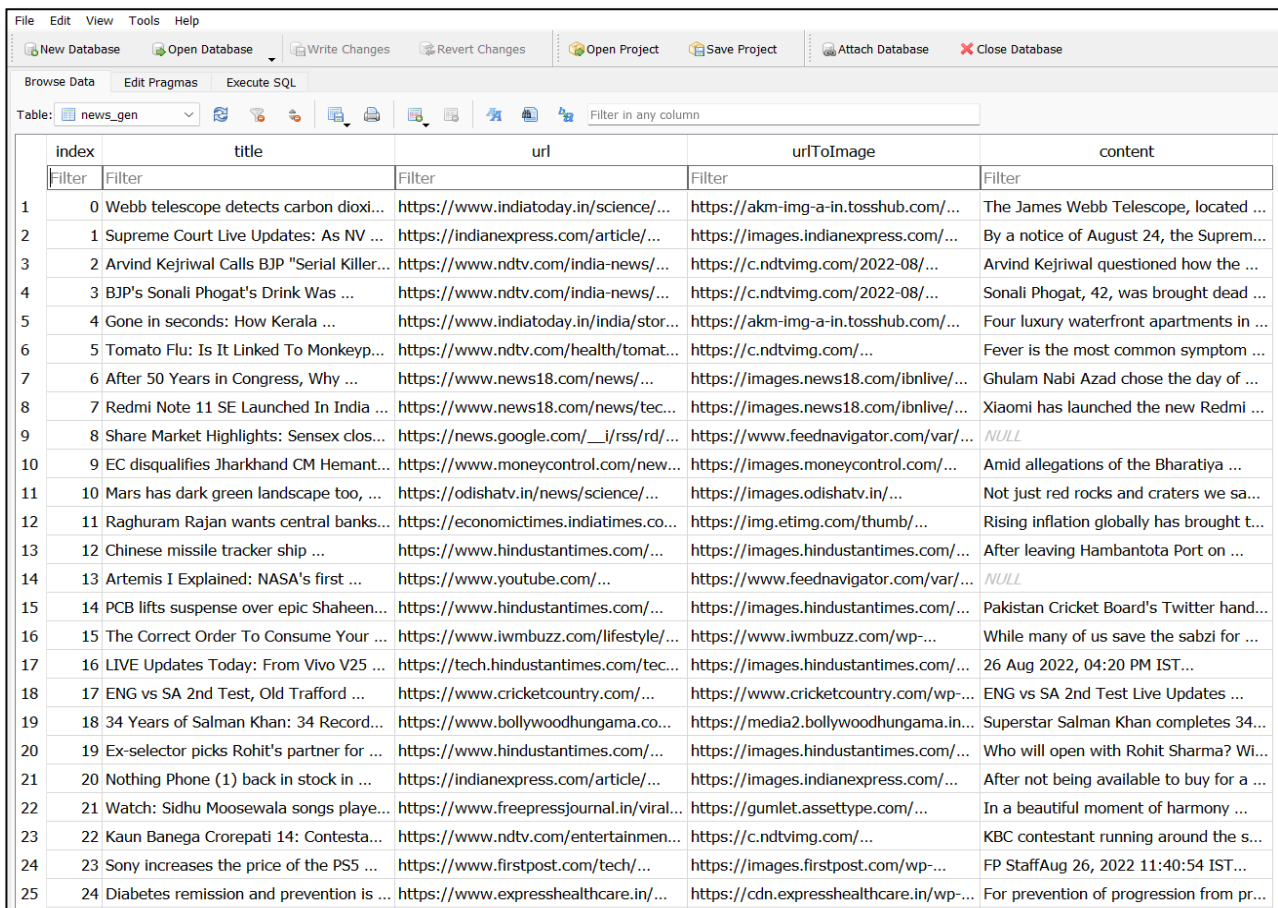
You can close the news application by clicking the close button in the top right corner. The news application will be closed.

TABLES CREATED

TABLE – (USER ACCOUNTS DATABASE): (PRIMARY KEY: ID)

id	name	email	password	phno
1	simeon	simeon@gmail.com	123@	7777778888
2	tony	tony@gmail.com	234tkinter	2347890986
3	sashwath	sashwath@yahoo.com	sashwath123:-)	9089907856
4	Abhinov	abhi@gmail.com	123^123	3457890009

TABLE – (NEWS DATABASE):



index	title	url	urlToImage	content
1	0 Webb telescope detects carbon diox...	https://www.indiatoday.in/science/...	https://akm-img-a-in.tosshub.com/...	The James Webb Telescope, located ...
2	1 Supreme Court Live Updates: As NV ...	https://indianexpress.com/article/...	https://images.indianexpress.com/...	By a notice of August 24, the Suprem...
3	2 Arvind Kejriwal Calls BJP "Serial Killer...	https://www.ndtv.com/india-news/...	https://c.ndtvimg.com/2022-08/...	Arvind Kejriwal questioned how the ...
4	3 BJP's Sonali Phogat's Drink Was ...	https://www.ndtv.com/india-news/...	https://c.ndtvimg.com/2022-08/...	Sonali Phogat, 42, was brought dead ...
5	4 Gone in seconds: How Kerala ...	https://www.indiatoday.in/india/stor...	https://akm-img-a-in.tosshub.com/...	Four luxury waterfront apartments in ...
6	5 Tomato Flu: Is It Linked To Monkeyp...	https://www.ndtv.com/health/tomat...	https://c.ndtvimg.com/...	Fever is the most common symptom ...
7	6 After 50 Years in Congress, Why ...	https://www.news18.com/news/tec...	https://images.news18.com/ibnlive/...	Ghulam Nabi Azad chose the day of ...
8	7 Redmi Note 11 SE Launched In India ...	https://www.news18.com/news/tec...	https://images.news18.com/ibnlive/...	Xiaomi has launched the new Redmi ...
9	8 Share Market Highlights: Sensex clos...	https://news.google.com/_i/rss/rd/...	https://www.feednavigator.com/var/...	NULL
10	9 EC disqualifies Jharkhand CM Hemant...	https://www.moneycontrol.com/new...	https://images.moneycontrol.com/...	Amid allegations of the Bharatiya ...
11	10 Mars has dark green landscape too, ...	https://odishatv.in/news/science/...	https://images.odishatv.in/...	Not just red rocks and craters we sa...
12	11 Raghuram Rajan wants central banks...	https://economictimes.indiatimes.co...	https://img.etimg.com/thumb/...	Rising inflation globally has brought t...
13	12 Chinese missile tracker ship ...	https://www.hindustantimes.com/...	https://images.hindustantimes.com/...	After leaving Hambantota Port on ...
14	13 Artemis I Explained: NASA's first ...	https://www.youtube.com/...	https://www.feednavigator.com/var/...	NULL
15	14 PCB lifts suspense over epic Shaheen...	https://www.hindustantimes.com/...	https://images.hindustantimes.com/...	Pakistan Cricket Board's Twitter hand...
16	15 The Correct Order To Consume Your ...	https://www.iwmbuzz.com/lifestyle/...	https://www.iwmbuzz.com/wp-...	While many of us save the sabzi for ...
17	16 LIVE Updates Today: From Vivo V25 ...	https://tech.hindustantimes.com/tec...	https://images.hindustantimes.com/...	26 Aug 2022, 04:20 PM IST...
18	17 ENG vs SA 2nd Test, Old Trafford ...	https://www.cricketcountry.com/...	https://www.cricketcountry.com/wp-...	ENG vs SA 2nd Test Live Updates ...
19	18 34 Years of Salman Khan: 34 Record...	https://www.bollywoodhungama.co...	https://media2.bollywoodhungama.in...	Superstar Salman Khan completes 34...
20	19 Ex-selector picks Rohit's partner for ...	https://www.hindustantimes.com/...	https://images.hindustantimes.com/...	Who will open with Rohit Sharma? Wi...
21	20 Nothing Phone (1) back in stock in ...	https://indianexpress.com/article/...	https://images.indianexpress.com/...	After not being available to buy for a ...
22	21 Watch: Sidhu Moosewala songs playe...	https://www.freepressjournal.in/viral...	https://gumlet.assettype.com/...	In a beautiful moment of harmony ...
23	22 Kaun Banega Crorepati 14: Contesta...	https://www.ndtv.com/entertainmen...	https://c.ndtvimg.com/...	KBC contestant running around the s...
24	23 Sony increases the price of the PS5 ...	https://www.firstpost.com/tech/...	https://images.firstpost.com/wp-...	FP StaffAug 26, 2022 11:40:54 IST...
25	24 Diabetes remission and prevention is ...	https://www.expresshealthcare.in/...	https://cdn.expresshealthcare.in/wp-...	For prevention of progression from pr...

The above table is used to store the data parsed from the JSON given by the API as response. It stores the titles, URL, URL to the image and the content of the news.

CONCLUSION

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The **NEWS SCOOPS** application is a very useful application for knowing what is happening around us. It is necessary for knowing the current affairs, thus the NEWS SCOOPS application will serve as tool for getting your news updates. This application is designed with ample foresight keeping the user in mind.

Some Important Features:

- ❖ Well-developed and simple to use GUI
- ❖ Highly user friendly
- ❖ Different categories of news at your fingertips
- ❖ Detailed News Updates
- ❖ News with headlines as well as pictures
- ❖ Timely updating News
- ❖ Time updating weather forecast

It has been designed in a way that you also get your weather updates also. It also redirects you to the news source if you click the read more option. It also gives a preview of the news. This application helps in getting your news updates easily and efficiently, without much effort or discomfort.

FUTURE ENHANCEMENTS

FUTURE ENHANCEMENTS

Development of this application in the future:

- ❖ In the future, this application can be enhanced by converting it into an independent application or website that can run without the terminal or Python shell.
- ❖ This application will be enhanced by adding a search feature for searching news.
- ❖ Feedback Window for the user to send feedback on the application.
- ❖ Like and Reaction Button will be added for the user to interact with the news application.
- ❖ Features to help the users provide ratings and reviews can also be added in the future enhancements. This will help in more interaction and user-friendliness.
- ❖ Features to view the history of the news which the user has viewed already can also be added. This will help the user to view the old news which he viewed.
- ❖ Features to manage interest of the news categories the user wants can also be added.
- ❖ More news categories can also be added to provide news on a diverse range.

BIBLIOGRAPHY

BIBLIOGRAPHY

- + Codemy YouTube: <https://www.youtube.com/c/Codemycom>
- + Google Search: <https://www.google.com/search>
- + GeeksforGeeks: <https://www.geeksforgeeks.org/>
- + News API: <https://newsapi.org/>
- + OpenWeatherMap API: <https://openweathermap.org/api>
- + Official Python Documentation: <https://docs.python.org/>
- + Stack Overflow: <https://stackoverflow.com/>
- + YouTube: <https://www.youtube.com/>
- + Various helpful online communities